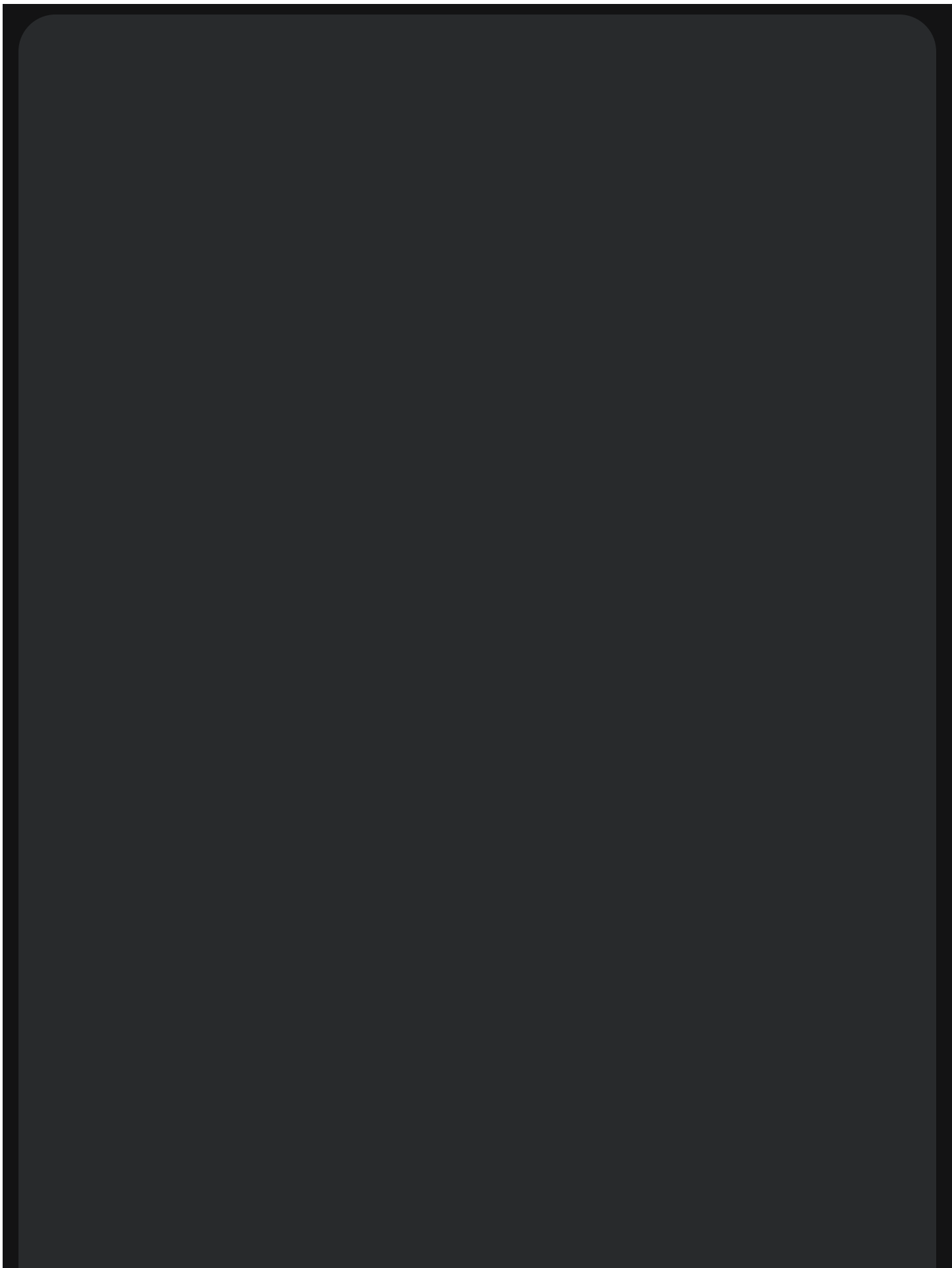




```
1 #Data analysis libraries
2 import numpy as np
3 import pandas as pd
4 pd.set_option('display.max_columns', 100) # Show all columns when
5 import warnings
6 warnings.filterwarnings('ignore')
7
8 # libraries for models
9 from sklearn.model_selection import train_test_split
10 from sklearn.model_selection import cross_val_score
11 from sklearn.ensemble import GradientBoostingClassifier, RandomForestClassifier
12 from sklearn.linear_model import LogisticRegression
13 from sklearn.svm import SVC
14 from sklearn.linear_model import LinearRegression
15 from sklearn.neighbors import KNeighborsClassifier
16 from sklearn.naive_bayes import GaussianNB
17 from sklearn.neural_network import MLPClassifier
18 from sklearn.tree import DecisionTreeClassifier
19 from pandas.core.common import random_state
20 from sklearn.model_selection import train_test_split, cross_val_score
21 from sklearn.model_selection import GridSearchCV
22 # metrics evaluation libraries
23 from sklearn.metrics import auc, classification_report, confusion_matrix
24 from sklearn.metrics import precision_recall_curve, precision_score, recall_score
25 from sklearn.model_selection import GridSearchCV, RandomizedSearchCV
26 #preprocess libraries
27 from sklearn.preprocessing import LabelEncoder
28 from sklearn.preprocessing import StandardScaler
29 from sklearn.decomposition import PCA
30 #RFE
31 from sklearn.feature_selection import RFE
32 #C-B
33 from sklearn.feature_selection import SelectKBest
34 from sklearn.feature_selection import f_regression
35 #F-importance
36 from sklearn.ensemble import RandomForestClassifier
37 #MUTUAL-info
38 from sklearn.feature_selection import mutual_info_classif
39 from sklearn.feature_selection import mutual_info_regression
40 #visulization libraries
41 import seaborn as sns
42 import matplotlib.pyplot as plt
43 import plotly.graph_objs as go
44 import plotly.offline as py
45 from plotly.offline import iplot, init_notebook_mode
46 from plotly.graph_objs import Scatter
47 py.init_notebook_mode(connected=True)
```

```
47 from random import choice, randint
48 from matplotlib import legend
49 import random
```

```
1 # Read CSV Data
2 data = pd.read_csv("/content/thyroidDF.csv")
```

```
1 data.head(10)
```

	age	sex	on_thyroxine	query_on_thyroxine	on_antithyroid_meds	sick	pre
0	29	F	f	f	f	f	f
1	29	F	f	f	f	f	f
2	41	F	f	f	f	f	f
3	36	F	f	f	f	f	f
4	32	F	f	f	f	f	f
5	60	F	f	f	f	f	f
6	77	F	f	f	f	f	f
7	28	F	f	f	f	f	f
8	28	F	f	f	f	f	f
9	28	F	f	f	f	f	f

```
1 data.shape
```

```
(9172, 31)
```

```
1 data.duplicated().sum()
```

```
0
```

```
1 data.info()
```

```
<class 'pandas.core.frame.DataFrame'>
Index: 7679 entries, 0 to 9171
Data columns (total 28 columns):
#   Column                Non-Null Count  Dtype
---  -
0   age                    7679 non-null   int64
1   sex                    7679 non-null   int64
2   on_thyroxine            7679 non-null   int64
```

```
3  query_on_thyroxine      7679 non-null  int64
4  on_antithyroid_meds    7679 non-null  int64
5  sick                    7679 non-null  int64
6  pregnant                7679 non-null  int64
7  thyroid_surgery         7679 non-null  int64
8  I131_treatment          7679 non-null  int64
9  query_hypothyroid       7679 non-null  int64
10 query_hyperthyroid      7679 non-null  int64
11 lithium                 7679 non-null  int64
12 goitre                  7679 non-null  int64
13 tumor                   7679 non-null  int64
14 hypopituitary           7679 non-null  int64
15 psych                   7679 non-null  int64
16 TSH                     7679 non-null  float64
17 T3                      7679 non-null  float64
18 TT4                     7679 non-null  float64
19 T4U                     7679 non-null  float64
20 FTI                     7679 non-null  float64
21 target                  7679 non-null  object
22 referral_source_STMW    7679 non-null  int64
23 referral_source_SVHC    7679 non-null  int64
24 referral_source_SVHD    7679 non-null  int64
25 referral_source_SVI     7679 non-null  int64
26 referral_source_WEST    7679 non-null  int64
27 referral_source_other   7679 non-null  int64
dtypes: float64(5), int64(22), object(1)
memory usage: 1.7+ MB
```

```
1 data.count()
```

```
age                7679
sex                7679
on_thyroxine       7679
query_on_thyroxine 7679
on_antithyroid_meds 7679
sick               7679
pregnant           7679
thyroid_surgery    7679
I131_treatment     7679
query_hypothyroid  7679
query_hyperthyroid 7679
lithium            7679
goitre             7679
tumor              7679
hypopituitary      7679
psych              7679
TSH                7679
T3                 7679
TT4                7679
T4U                7679
FTI                7679
target             7679
referral_source_STMW 7679
referral_source_SVHC 7679
```

```
referral_source_SVHD    7679
referral_source_SVI     7679
referral_source_WEST    7679
referral_source_other   7679
dtype: int64
```

```
1 data.isna().sum()
```

```
age                0
sex                0
on_thyroxine       0
query_on_thyroxine 0
on_antithyroid_meds 0
sick               0
pregnant           0
thyroid_surgery    0
I131_treatment     0
query_hypothyroid  0
query_hyperthyroid 0
lithium            0
goitre             0
tumor              0
hypopituitary      0
psych              0
TSH                0
T3                 0
TT4                0
T4U                0
FTI                0
target             0
referral_source_STMW 0
referral_source_SVHC 0
referral_source_SVHD 0
referral_source_SVI 0
referral_source_WEST 0
referral_source_other 0
dtype: int64
```

```
1 data.isnull().sum()
```

```
age                0
sex                0
on_thyroxine       0
query_on_thyroxine 0
on_antithyroid_meds 0
sick               0
pregnant           0
thyroid_surgery    0
I131_treatment     0
query_hypothyroid  0
query_hyperthyroid 0
lithium            0
goitre             0
tumor              0
```

```

hypopituitary      0
psych              0
TSH                0
T3                0
TT4               0
T4U               0
FTI               0
target            0
referral_source_STMW 0
referral_source_SVHC 0
referral_source_SVHD 0
referral_source_SVI 0
referral_source_WEST 0
referral_source_other 0
dtype: int64

```

```
1 data = data.drop(['TSH_measured', 'T3_measured', 'TT4_measured', 'T4U_
```

```
1 data.describe(include='O').T
```

	count	unique	top	freq
target	7679	3	Negative	6771

```
1 data = pd.get_dummies(data, columns=['referral_source'])
```

```

1 num_col = ['age', 'TSH', 'T3', 'TT4', 'T4U', 'FTI', 'target']
2 cat_col = [col for col in data.columns if col not in num_col]
3 cat_col

```

```

['sex',
 'on_thyroxine',
 'query_on_thyroxine',
 'on_antithyroid_meds',
 'sick',
 'pregnant',
 'thyroid_surgery',
 'I131_treatment',
 'query_hypothyroid',
 'query_hyperthyroid',
 'lithium',
 'goitre',
 'tumor',
 'hypopituitary',
 'psych',
 'referral_source_STMW',
 'referral_source_SVHC',
 'referral_source_SVHD',
 'referral_source_SVI',
 'referral_source_WEST',

```

```
'referral_source_other']
```

```
1 lenc = LabelEncoder()
2 for i in cat_col:
3     data[i] = lenc.fit_transform(data[i])
```

```
1 data.head()
```

	age	sex	on_thyroxine	query_on_thyroxine	on_antithyroid_meds	sick	pre
0	29	0	0	0	0	0	
1	29	0	0	0	0	0	
2	41	0	0	0	0	0	
3	36	0	0	0	0	0	
4	32	0	0	0	0	0	

```
1 data['target'].unique()
```

```
array(['Negative', 'Hypothyroid', 'Hyperthyroid'], dtype=object)
```

```
1 map = {'-':"Negative", 'A':"Hyperthyroid", 'AK':"Hyperthyroid", 'B':"
2 'C':"Hyperthyroid", 'C|I': 'Hyperthyroid', 'D':"Hyperthyroid",
3 'D|R':"Hyperthyroid", 'E': "Hypothyroid", 'F': "Hypothyroid",
4 'FK': "Hypothyroid", "G": "Hypothyroid", "GK": "Hypothyroid",
5 "GI": "Hypothyroid", 'GKJ': 'Hypothyroid', 'H|K': 'Hypothyroid',
6 }
7 data['target'] = data['target'].map(map)
8 data.dropna(subset=['target'], inplace=True)
```

```
1 nul_co = ['TSH', 'T3', 'TT4', 'T4U', 'FTI' ]
```

✓ filling the Nan values with knnimputer

```
1 from sklearn.impute import SimpleImputer, KNNImputer
2 imputer_constant = SimpleImputer(strategy='constant', fill_value=0)
3 data[nul_co] = imputer_constant.fit_transform(data[nul_co])
```

✓ Save new dataset

```
1 data_new = data.to_csv('new_thyroid.csv')
```

```
1 data_new = pd.read_csv('/content/new_thyroid.csv')
2
```

```
1 data_new = data_new.drop(['Unnamed: 0'], axis = 1)
```

```
1 data_new.head()
```

	age	sex	on_thyroxine	query_on_thyroxine	on_antithyroid_meds	sick	preg
0	29	0	0	0	0	0	
1	29	0	0	0	0	0	
2	41	0	0	0	0	0	
3	36	0	0	0	0	0	
4	60	0	0	0	0	0	

```
1 data_new.shape
```

```
(7679, 28)
```

```
1 data_new.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 7679 entries, 0 to 7678
Data columns (total 28 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   age                                    7679 non-null   int64
1   sex                                    7679 non-null   float64
2   on_thyroxine                          7679 non-null   int64
3   query_on_thyroxine                    7679 non-null   int64
4   on_antithyroid_meds                   7679 non-null   int64
5   sick                                  7679 non-null   int64
6   pregnant                              7679 non-null   int64
7   thyroid_surgery                       7679 non-null   int64
8   I131_treatment                        7679 non-null   int64
9   query_hypothyroid                     7679 non-null   int64
10  query_hyperthyroid                     7679 non-null   int64
11  lithium                                7679 non-null   int64
12  goitre                                 7679 non-null   int64
13  tumor                                  7679 non-null   int64
14  hypopituitary                          7679 non-null   int64
```



```
15  psych                7679 non-null    int64
16  TSH                  7679 non-null    float64
17  T3                   7679 non-null    float64
18  TT4                  7679 non-null    float64
19  T4U                  7679 non-null    float64
20  FTI                  7679 non-null    float64
21  target               7679 non-null    int64
22  referral_source_STMW 7679 non-null    int64
23  referral_source_SVHC 7679 non-null    int64
24  referral_source_SVHD 7679 non-null    int64
25  referral_source_SVI  7679 non-null    int64
26  referral_source_WEST 7679 non-null    int64
27  referral_source_other 7679 non-null    int64
dtypes: float64(6), int64(22)
memory usage: 1.6 MB
```

```
1 data_new.isnull().sum()
```

```
age                0
sex                0
on_thyroxine       0
query_on_thyroxine 0
on_antithyroid_meds 0
sick               0
pregnant           0
thyroid_surgery    0
I131_treatment     0
query_hypothyroid  0
query_hyperthyroid 0
lithium            0
goitre             0
tumor              0
hypopituitary      0
psych              0
TSH                0
T3                 0
TT4                0
T4U                0
FTI                0
target             0
referral_source_STMW 0
referral_source_SVHC 0
referral_source_SVHD 0
referral_source_SVI 0
referral_source_WEST 0
referral_source_other 0
dtype: int64
```

```
1 data_new.isna().sum()
```

```
age                0
sex                0
on_thyroxine       0
query on thvroxine 0
```

```

on_antithyroid_meds    0
sick                   0
pregnant               0
thyroid_surgery        0
I131_treatment         0
query_hypothyroid      0
query_hyperthyroid     0
lithium                0
goitre                 0
tumor                  0
hypopituitary          0
psych                  0
TSH                    0
T3                     0
TT4                    0
T4U                    0
FTI                    0
target                 0
referral_source_STMW   0
referral_source_SVHC   0
referral_source_SVHD   0
referral_source_SVI    0
referral_source_WEST   0
referral_source_other  0
dtype: int64

```

```
1 data_new.describe()
```

	age	sex	on_thyroxine	query_on_thyroxine	on_antithyr
count	7679.000000	7679.000000	7679.000000	7679.000000	7679.000000
mean	52.044277	0.380909	0.110691	0.016018	0.000000
std	18.699990	0.549555	0.313770	0.125552	0.000000
min	1.000000	0.000000	0.000000	0.000000	0.000000
25%	37.000000	0.000000	0.000000	0.000000	0.000000
50%	55.000000	0.000000	0.000000	0.000000	0.000000
75%	67.000000	1.000000	0.000000	0.000000	0.000000
max	112.000000	2.000000	1.000000	1.000000	0.000000

```
1 data_new.count()
```

```

age                7679
sex                7679
on_thyroxine       7679
query_on_thyroxine 7679
on antithyroid meds 7679

```

```
sick 7679
pregnant 7679
thyroid_surgery 7679
I131_treatment 7679
query_hypothyroid 7679
query_hyperthyroid 7679
lithium 7679
goitre 7679
tumor 7679
hypopituitary 7679
psych 7679
TSH 7679
T3 7679
TT4 7679
T4U 7679
FTI 7679
target 7679
referral_source_STMW 7679
referral_source_SVHC 7679
referral_source_SVHD 7679
referral_source_SVI 7679
referral_source_WEST 7679
referral_source_other 7679
dtype: int64
```

▼ Detecting outliers

```
1 #finding skew of data
2 dskew=data_new.drop(['target'],axis=1)
3 pd.DataFrame(dskew.skew())
```

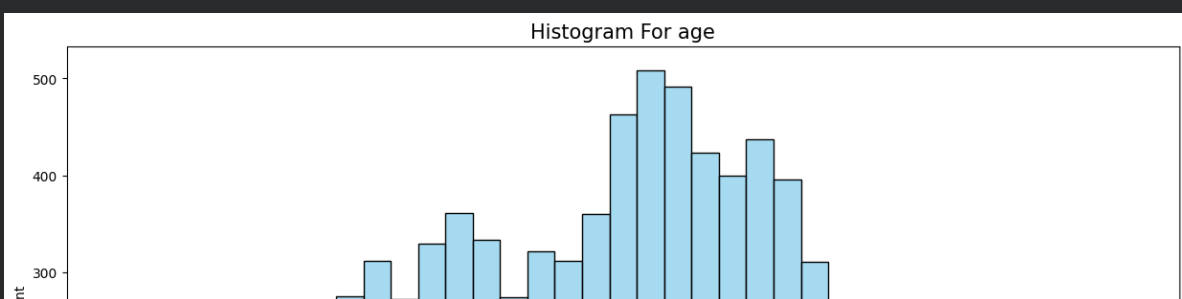
	0
age	50.556648
sex	1.079125
on_thyroxine	2.482135
query_on_thyroxine	7.711706
on_antithyroid_meds	8.972791
sick	4.878957
pregnant	13.749694
thyroid_surgery	8.335739
I131_treatment	7.176337
query_hypothyroid	3.441594

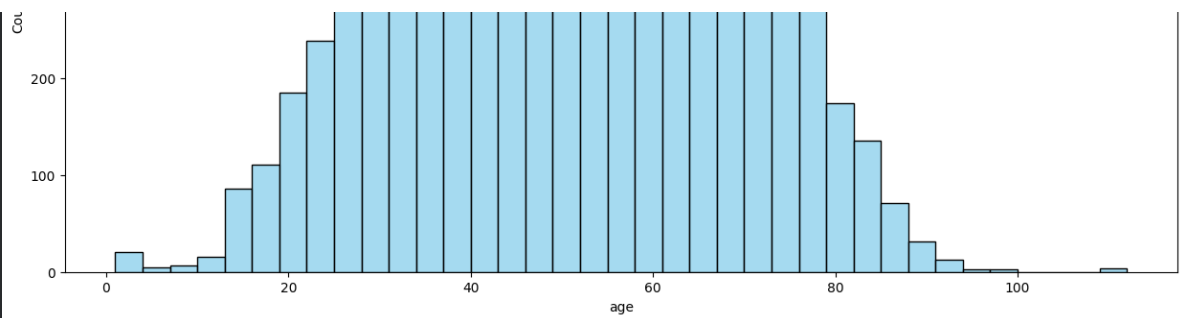
query_hypothyroid	3.281932
lithium	9.291703
goitre	10.040886
tumor	5.904944
hypopituitary	0.000000
psych	4.148950
TSH	12.938851
T3	1.103359
TT4	0.214787
T4U	-1.621319
FTI	1.084960
referral_source_STMW	6.944933
referral_source_SVHC	2.444698
referral_source_SVHD	11.182131
referral_source_SVI	1.167612
referral_source_WEST	61.951591
referral_source_other	-0.457212

```

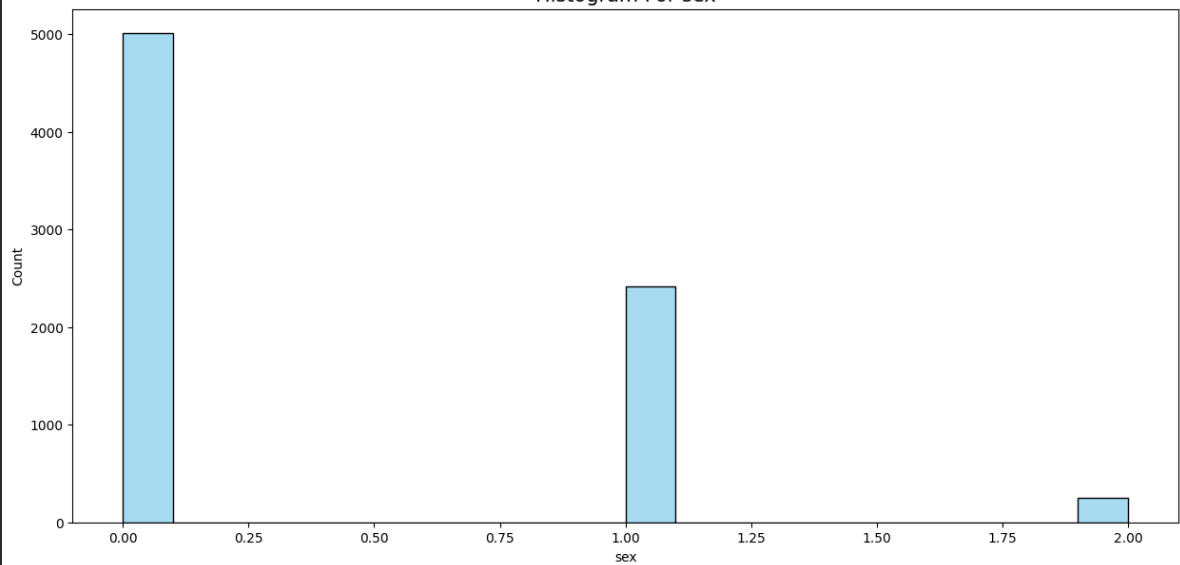
1 from matplotlib import pyplot as plt
2 num_var=[var for var in data_new.columns if var not in ["target"]]
3
4 # Plotting Box and Distribution plot
5 for var in num_var:
6     plt.figure(figsize=(15,7))
7     sns.histplot(data_new[var], kde=False, color='skyblue')
8     plt.title(f'Histogram For {var}', size=15)
9     plt.show()
10    ax.set_title(f'{var}')
11    ax.set_ylabel(var)
12    # plt.gca().spines[['top', 'right']].set_visible(False)

```

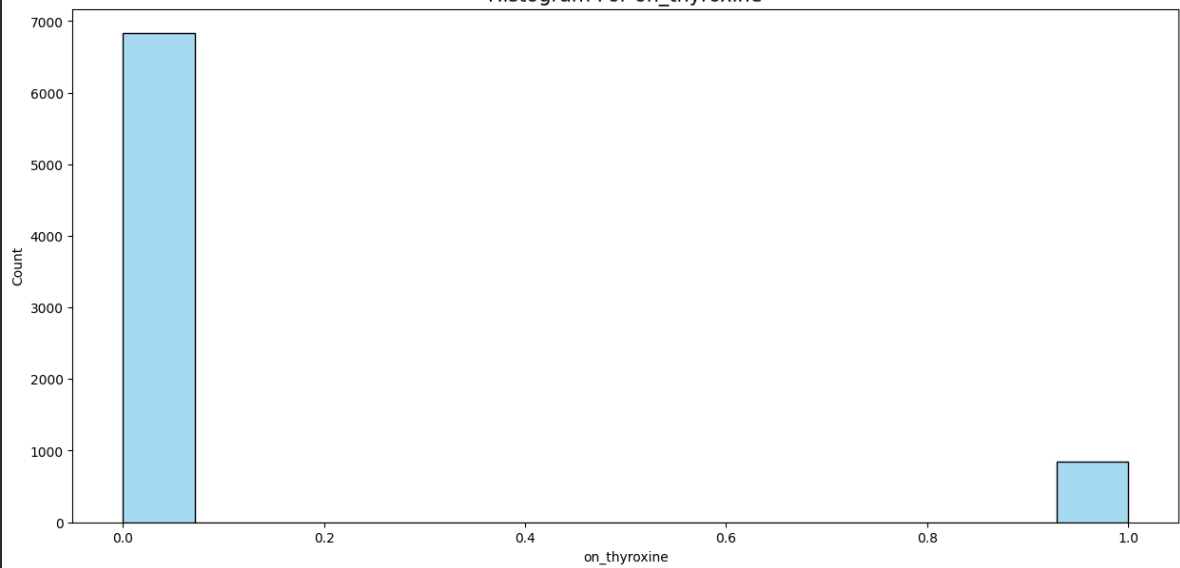




Histogram For sex

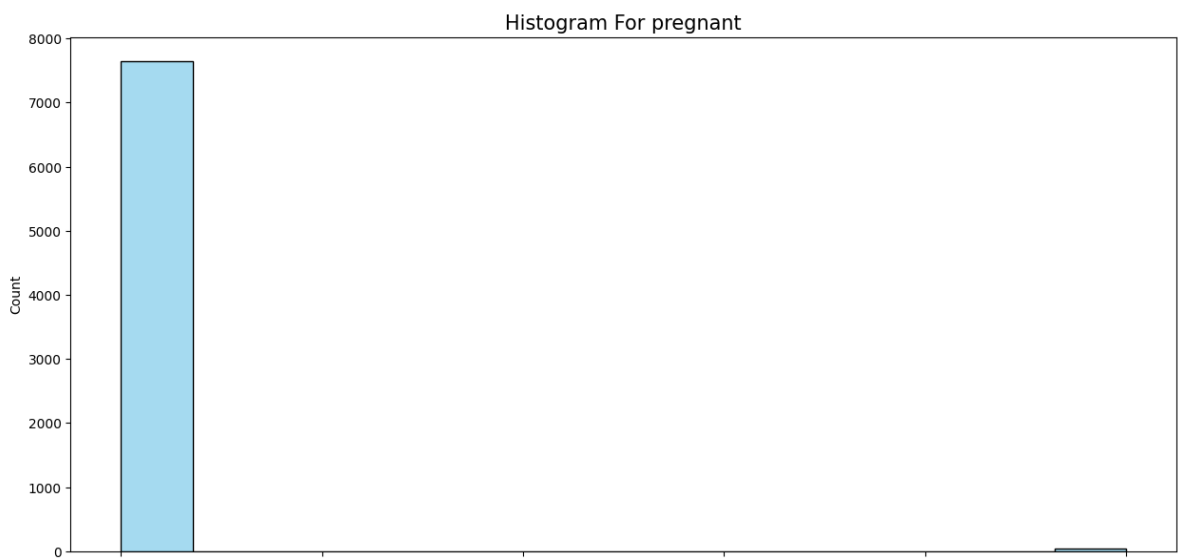
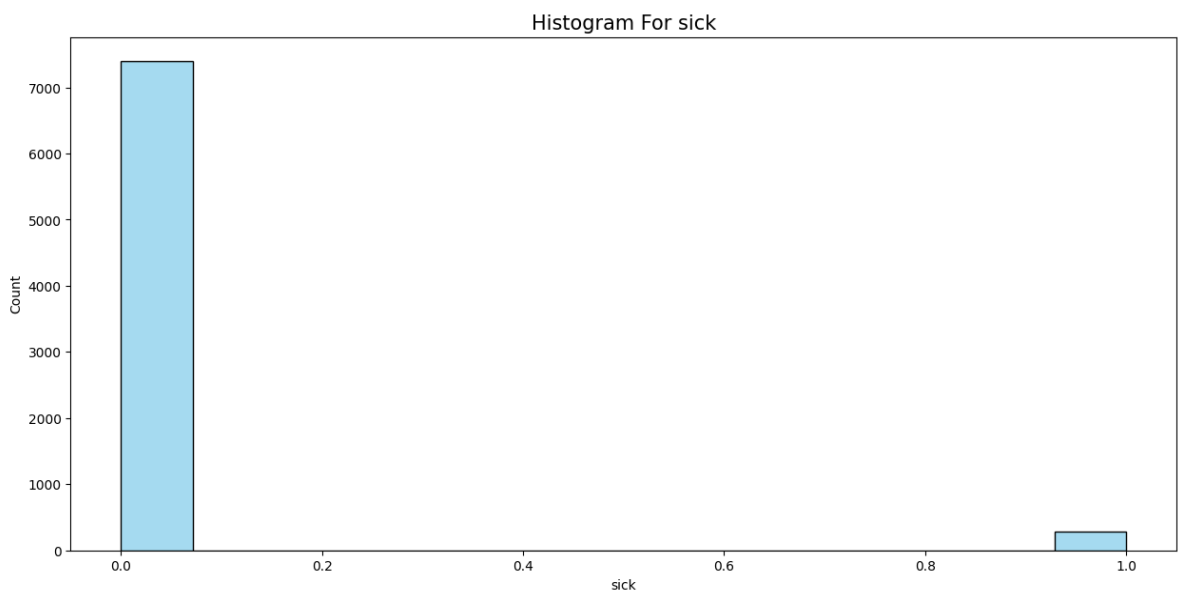
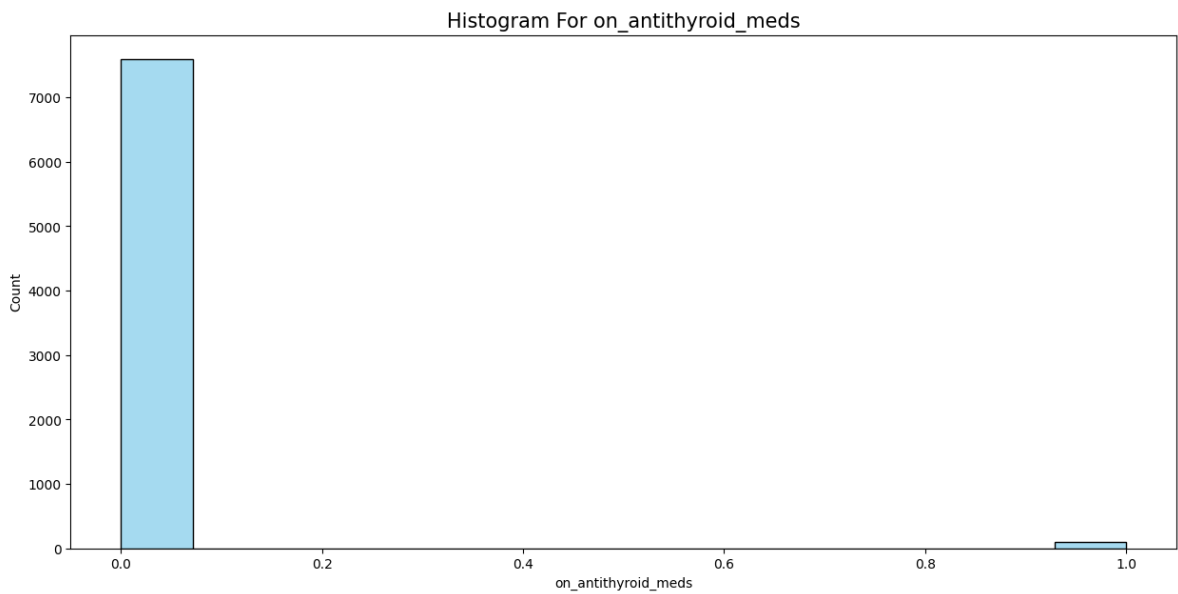
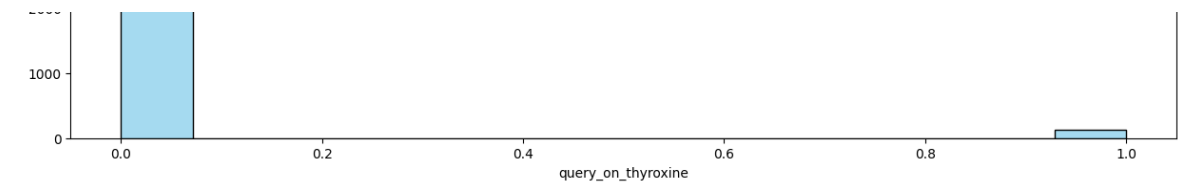


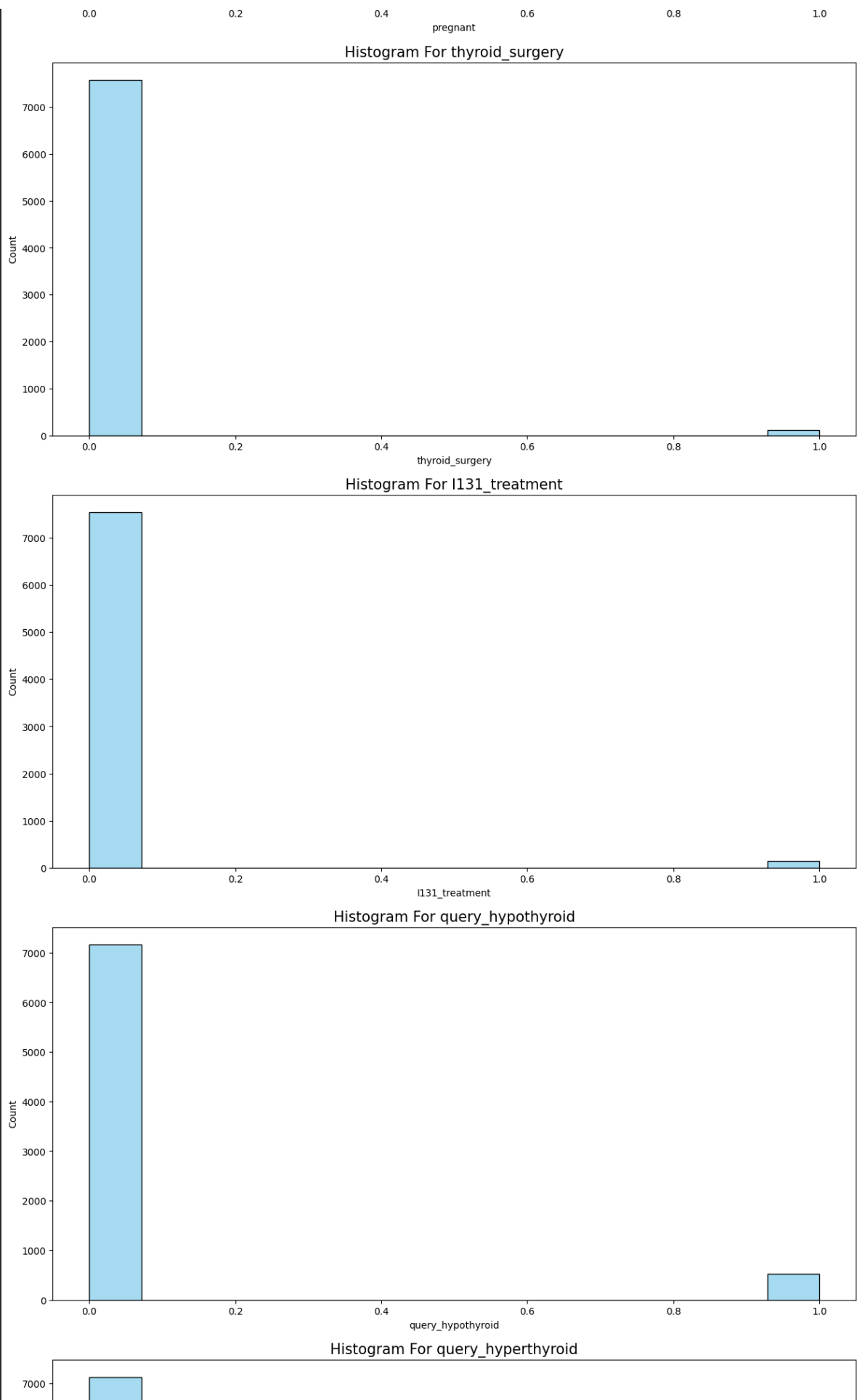
Histogram For on_thyroxine

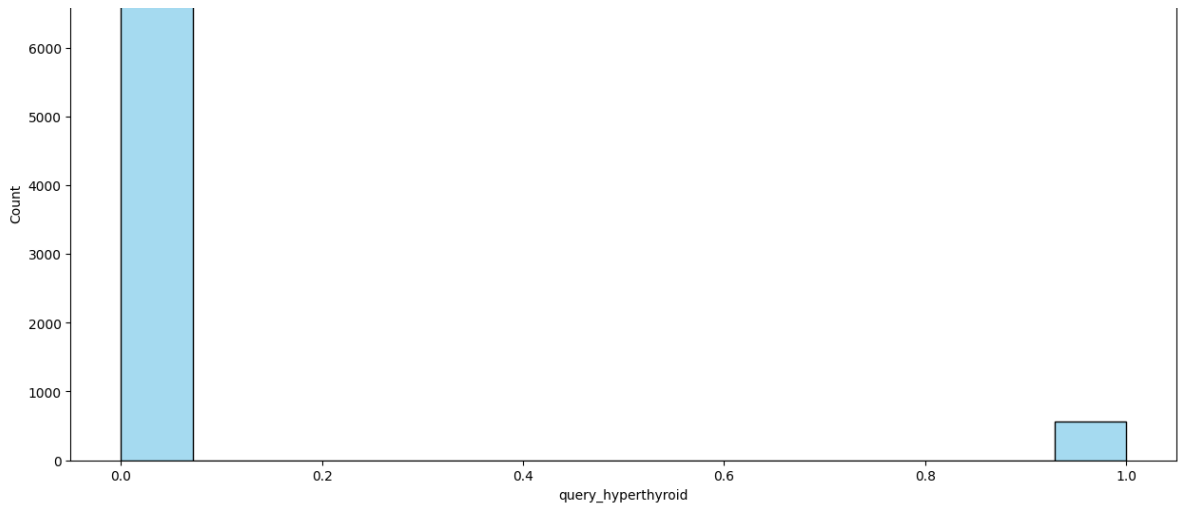


Histogram For query_on_thyroxine

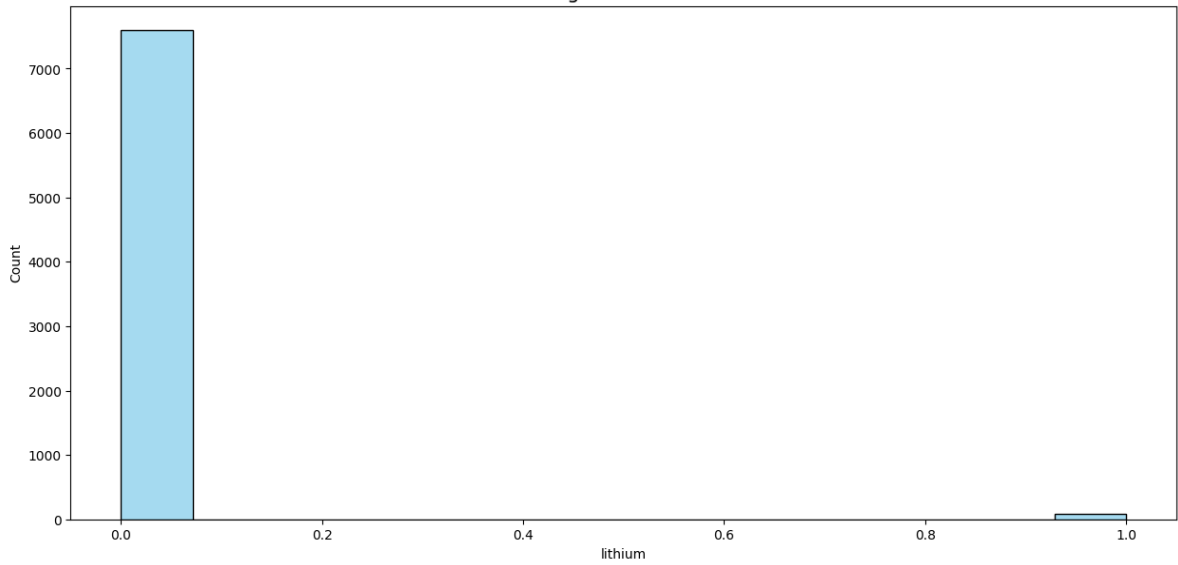




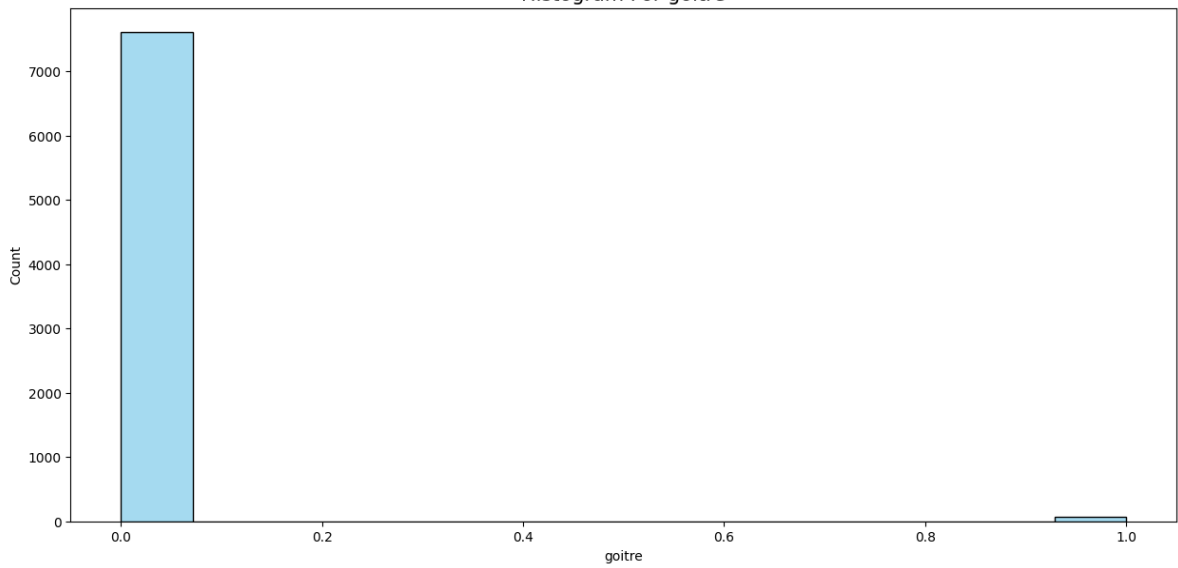




Histogram For lithium

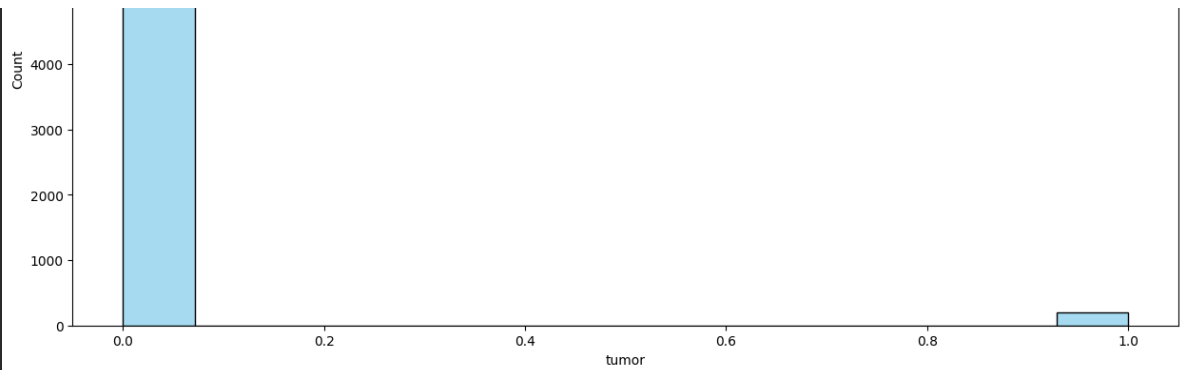


Histogram For goitre

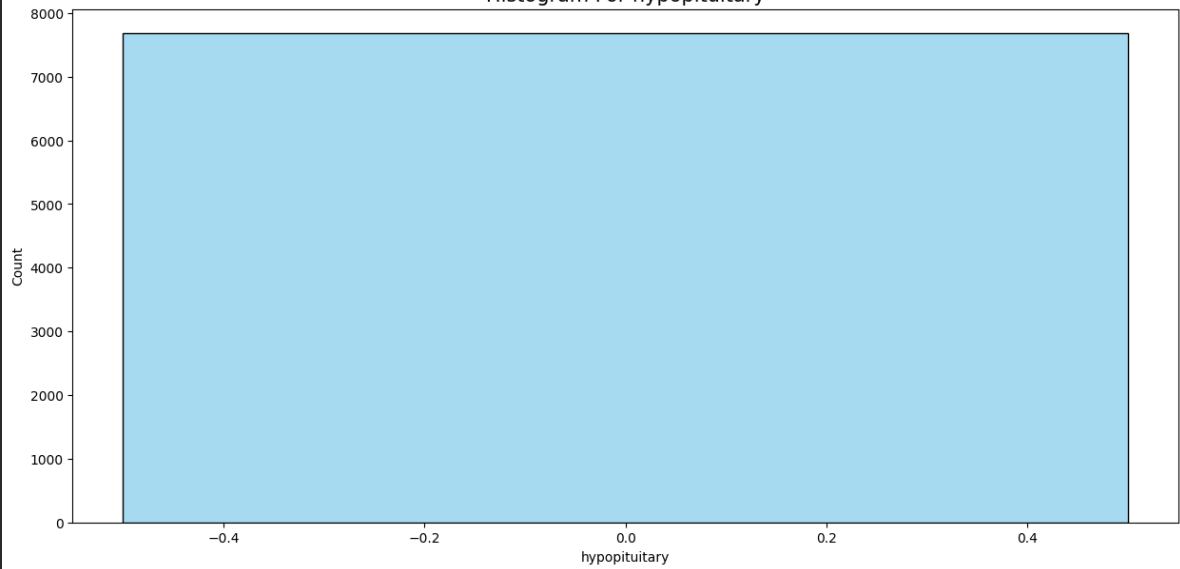


Histogram For tumor

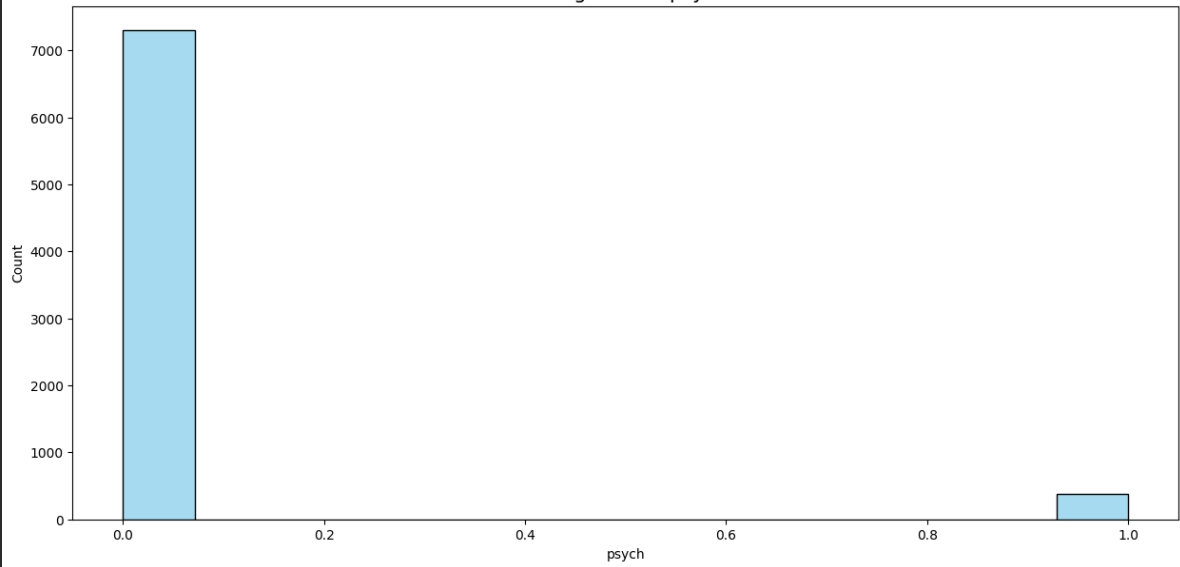




Histogram For hypopituitary

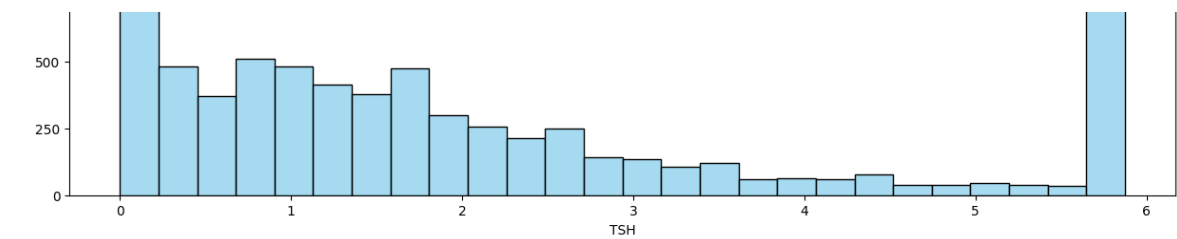


Histogram For psych

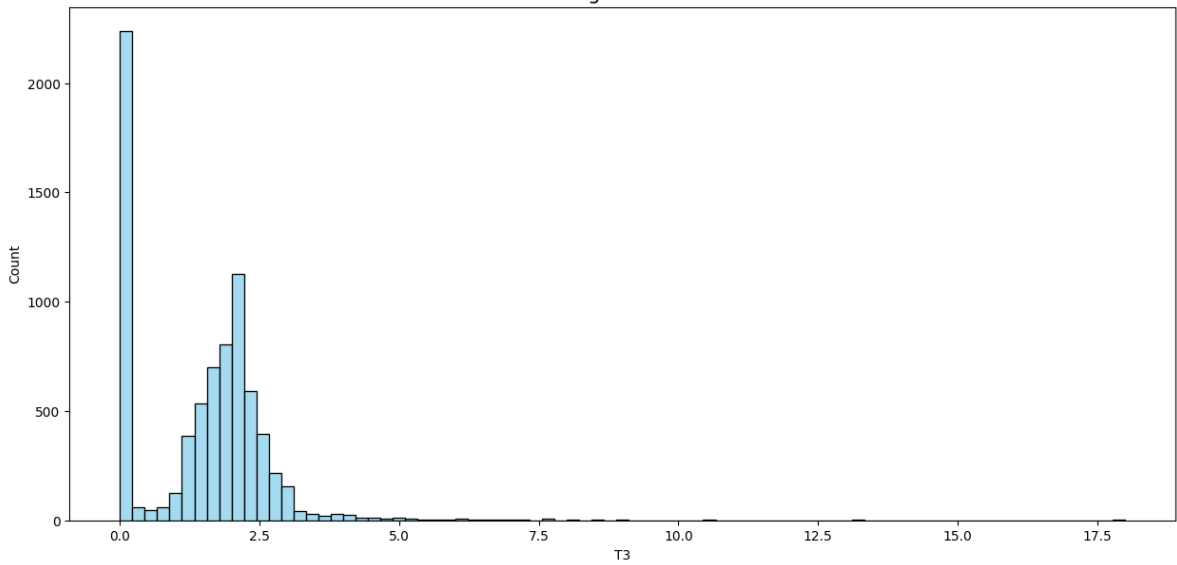


Histogram For TSH

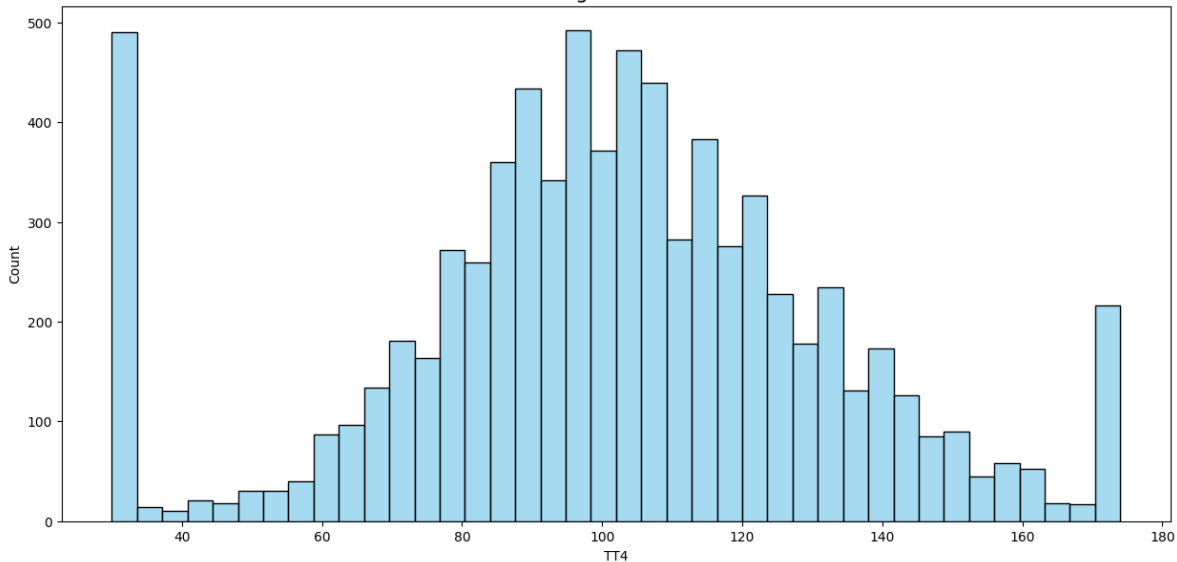




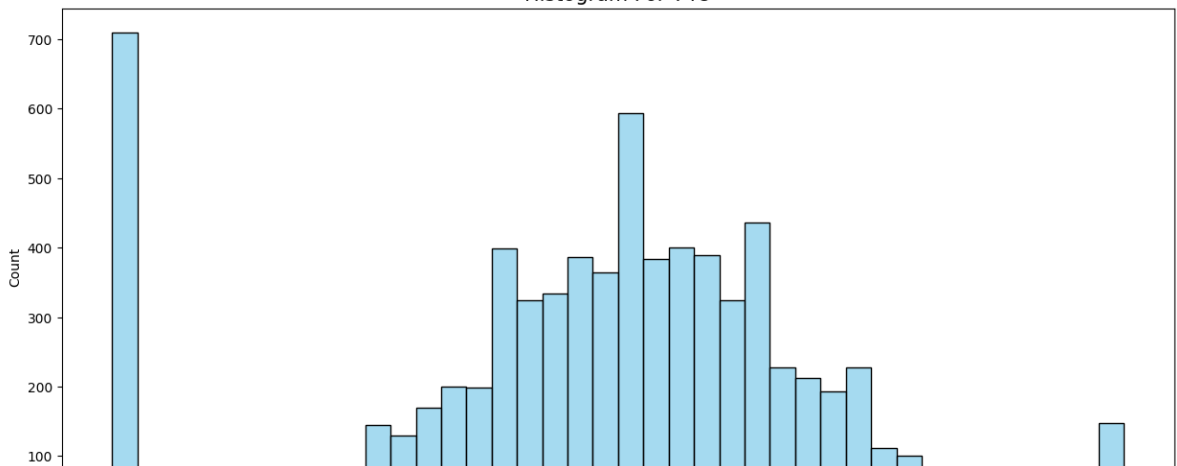
Histogram For T3

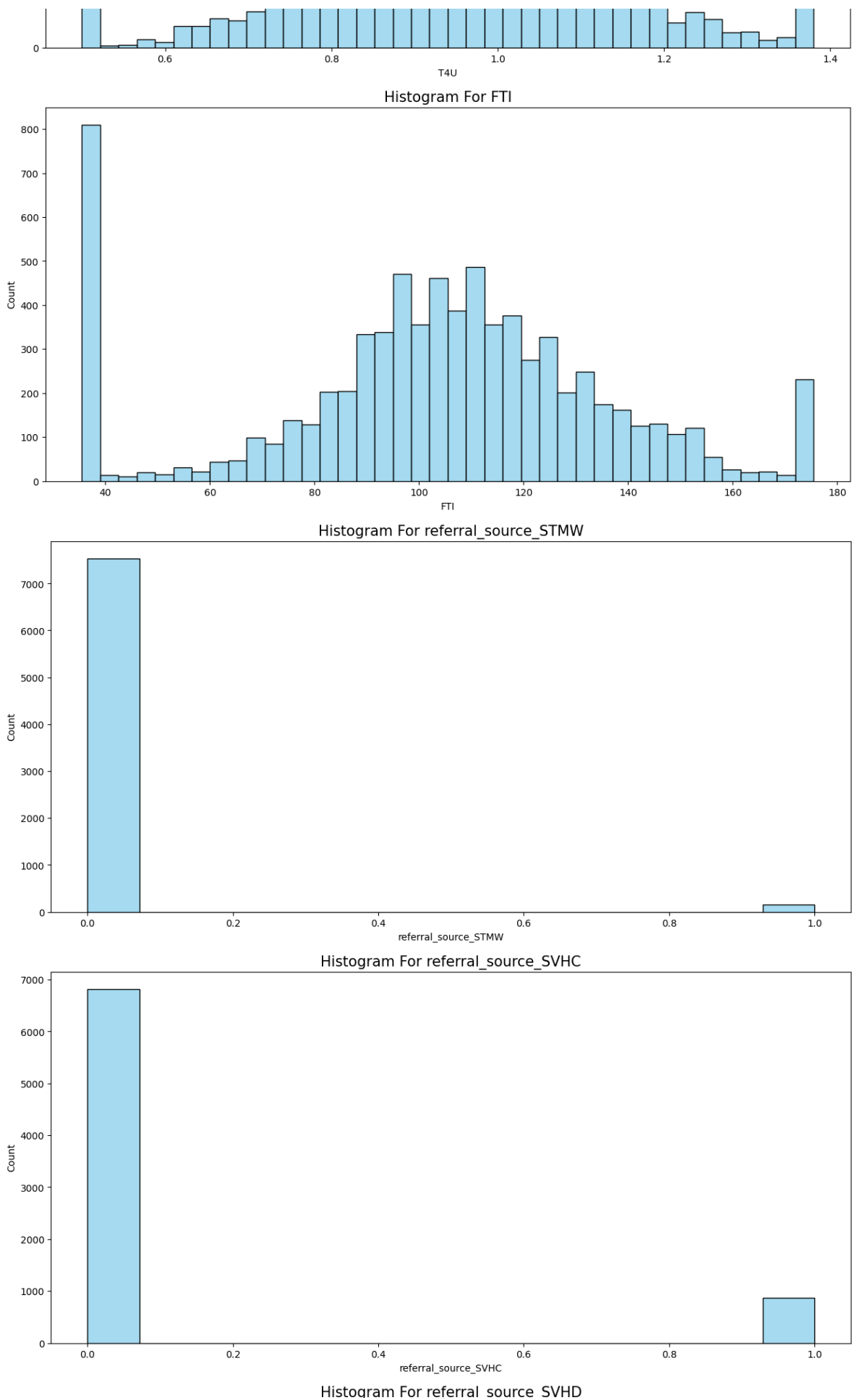


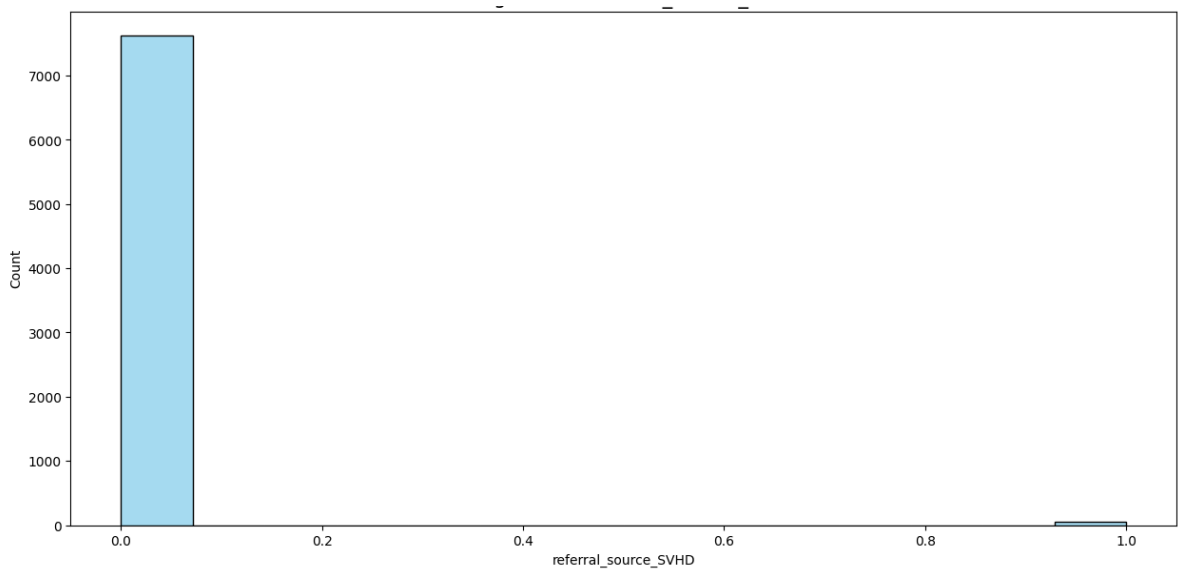
Histogram For TT4



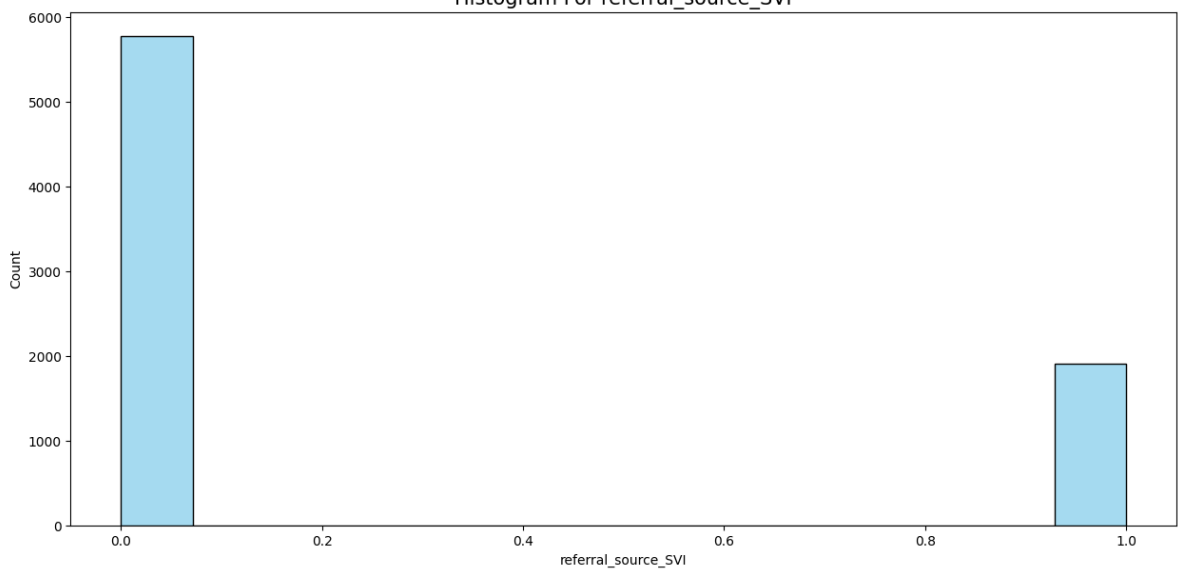
Histogram For T4U



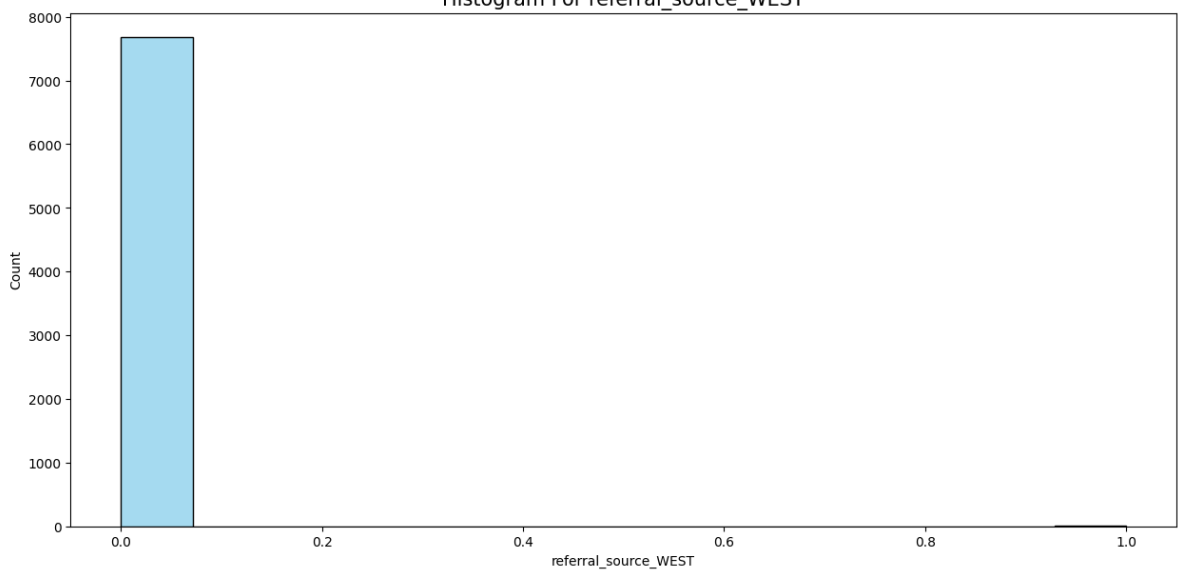




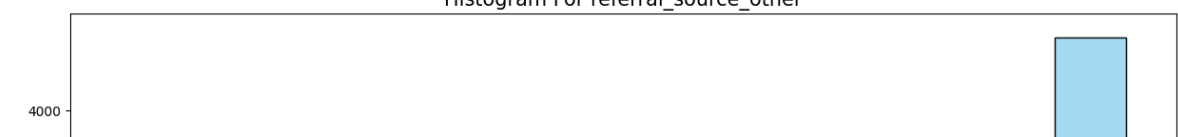
Histogram For referral_source_SVI

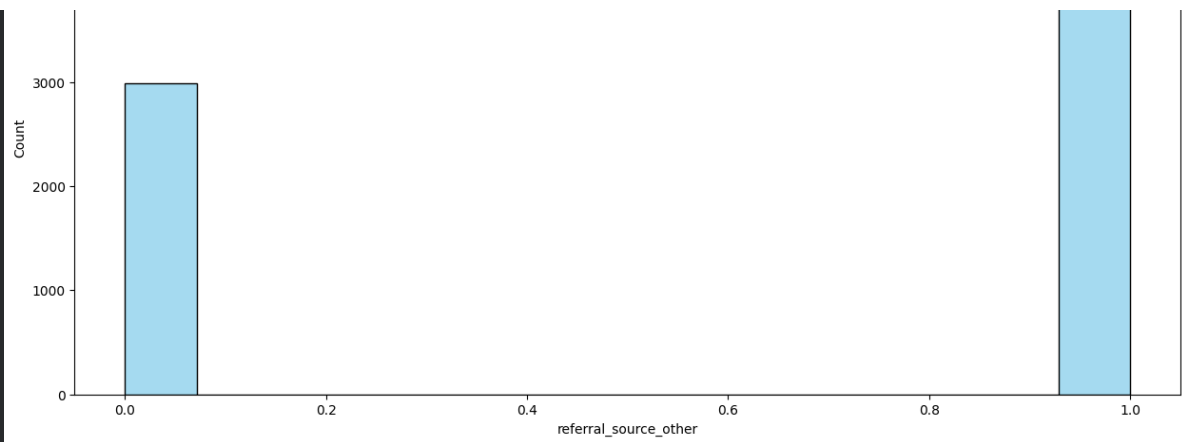


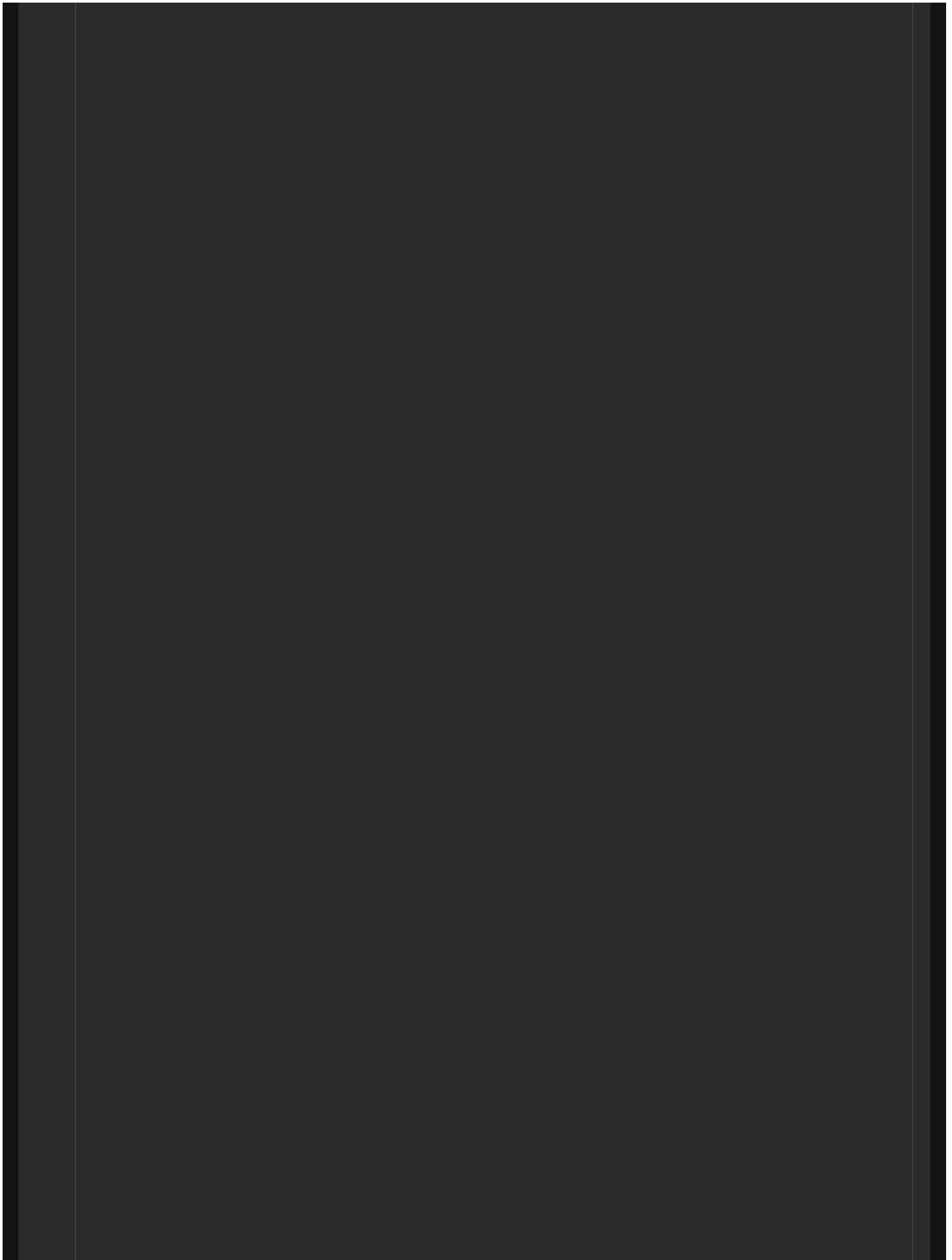
Histogram For referral_source_WEST

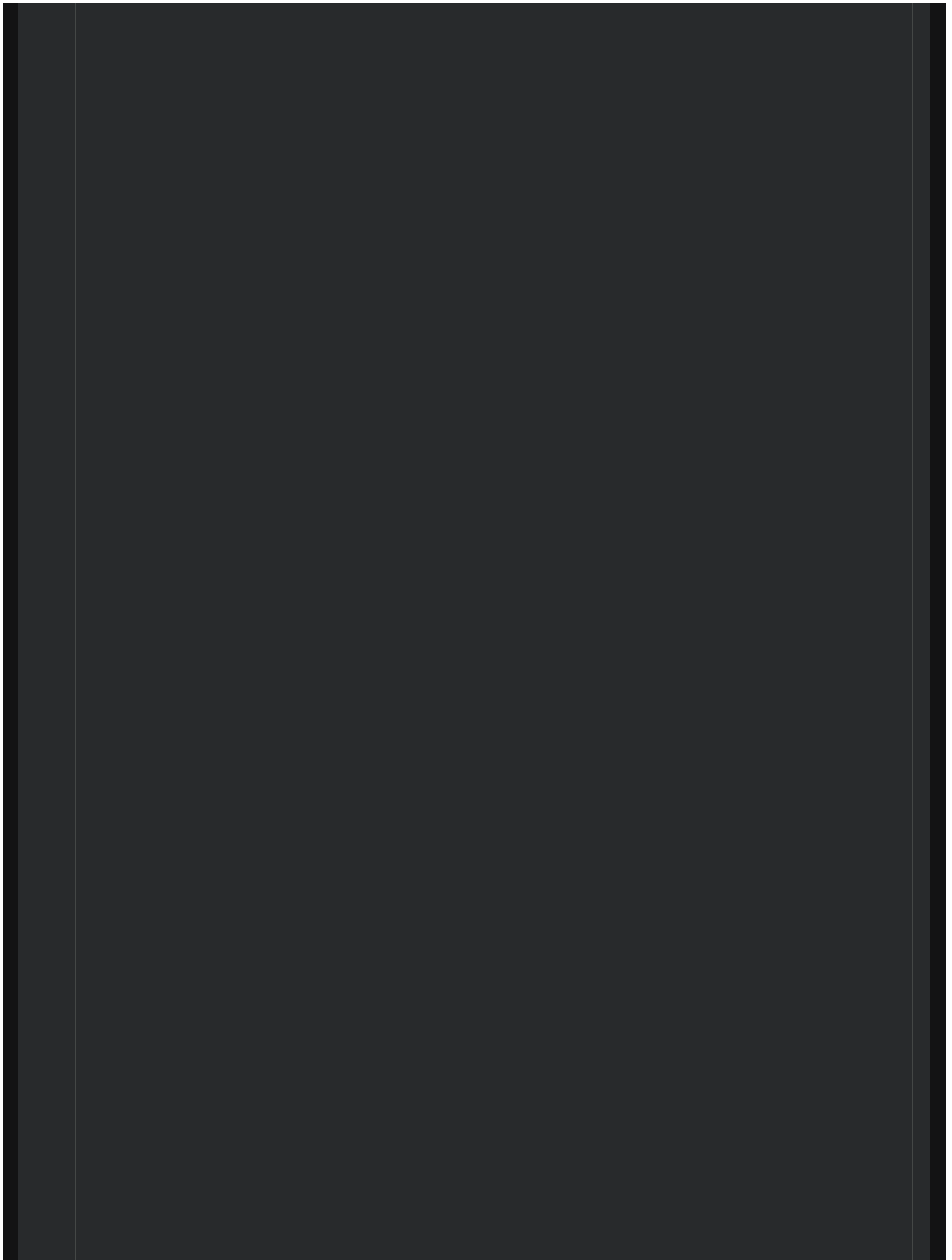


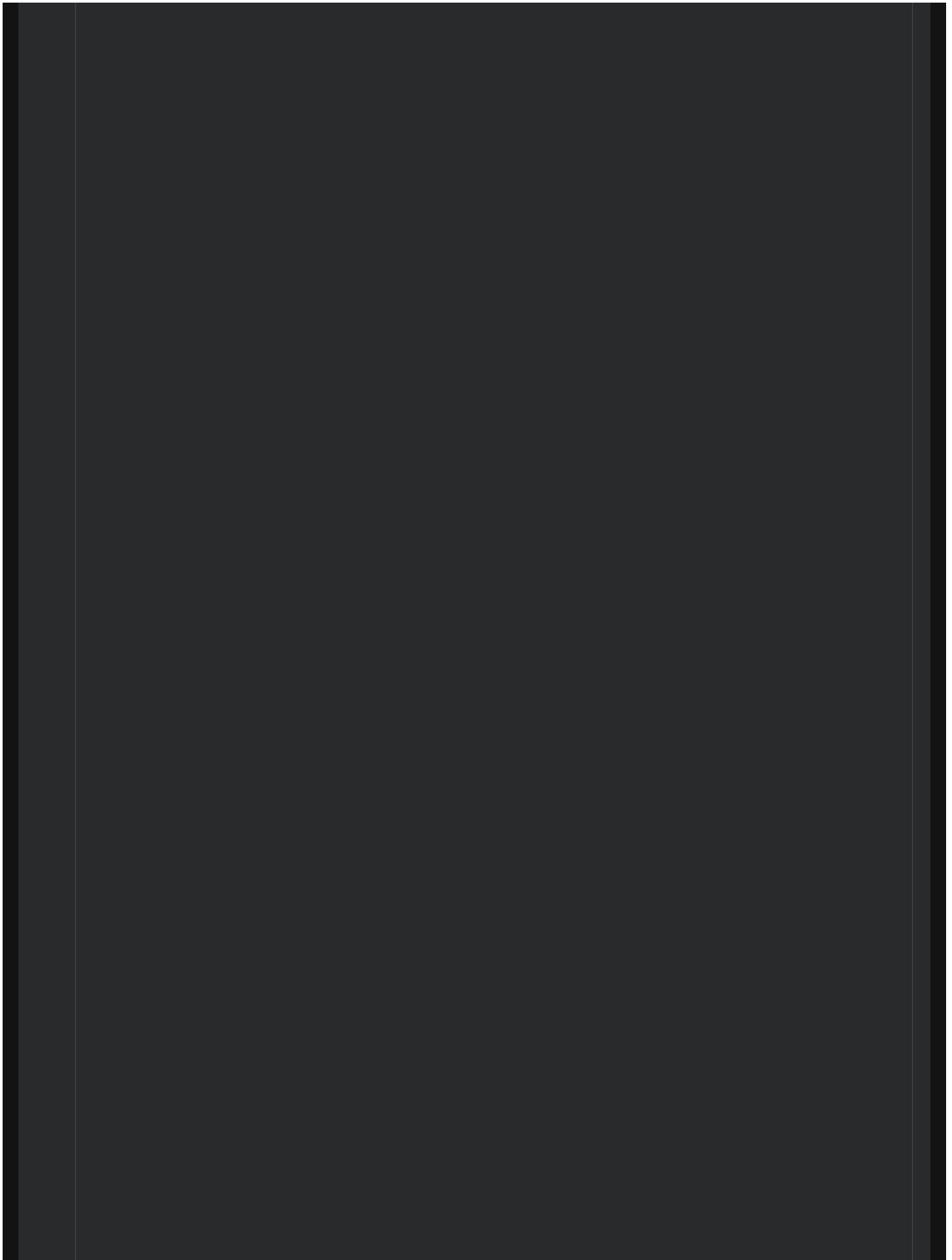
Histogram For referral_source_other

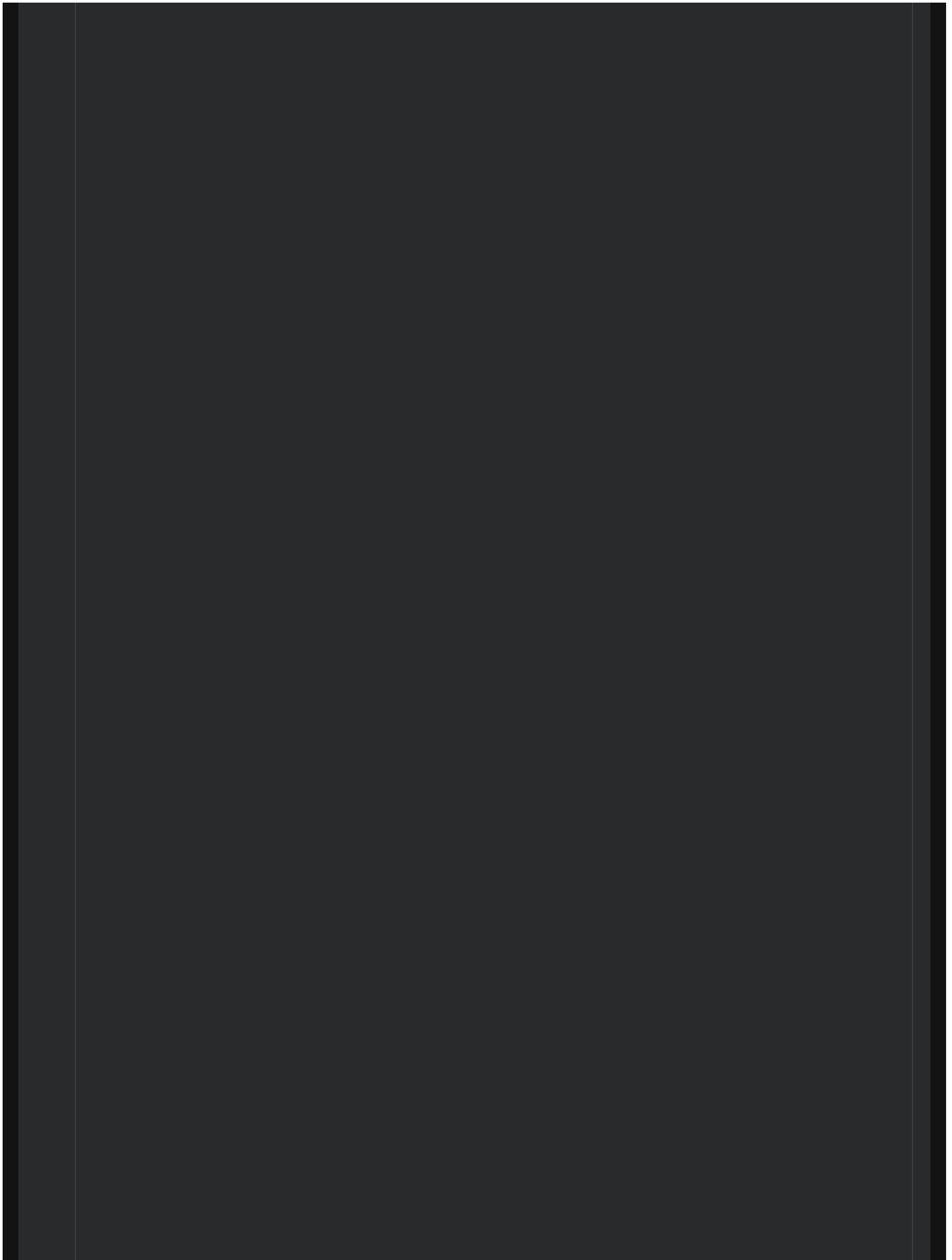


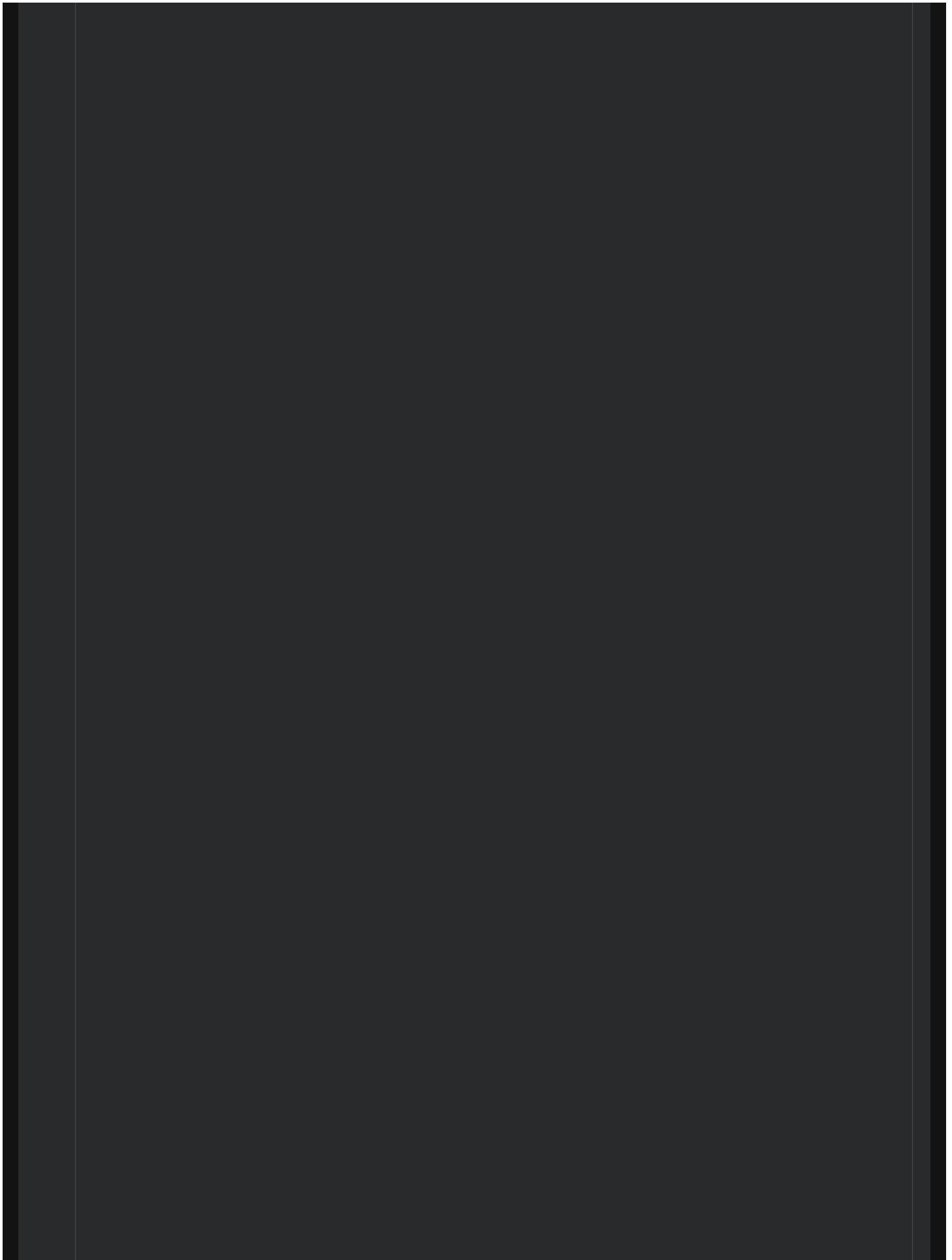


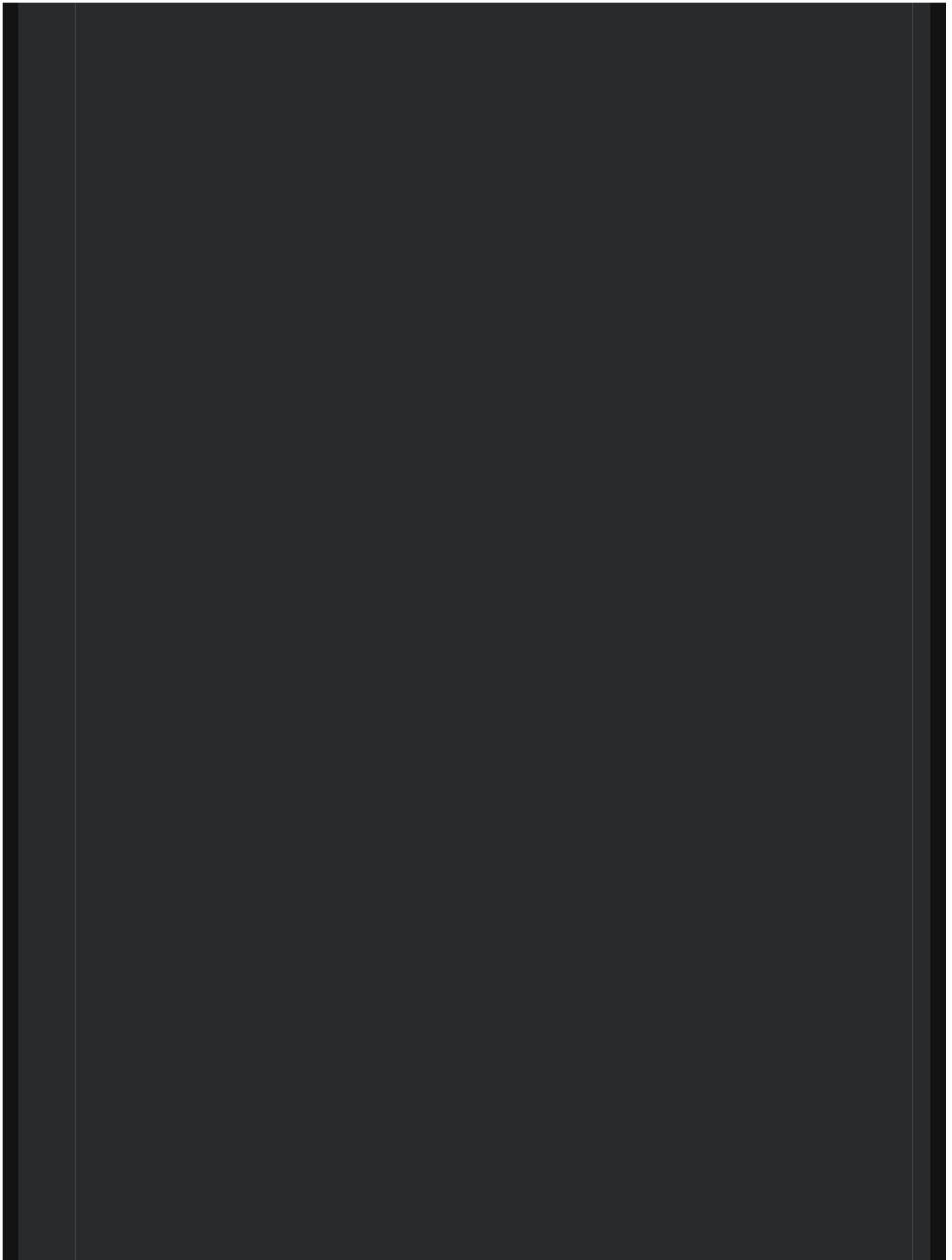


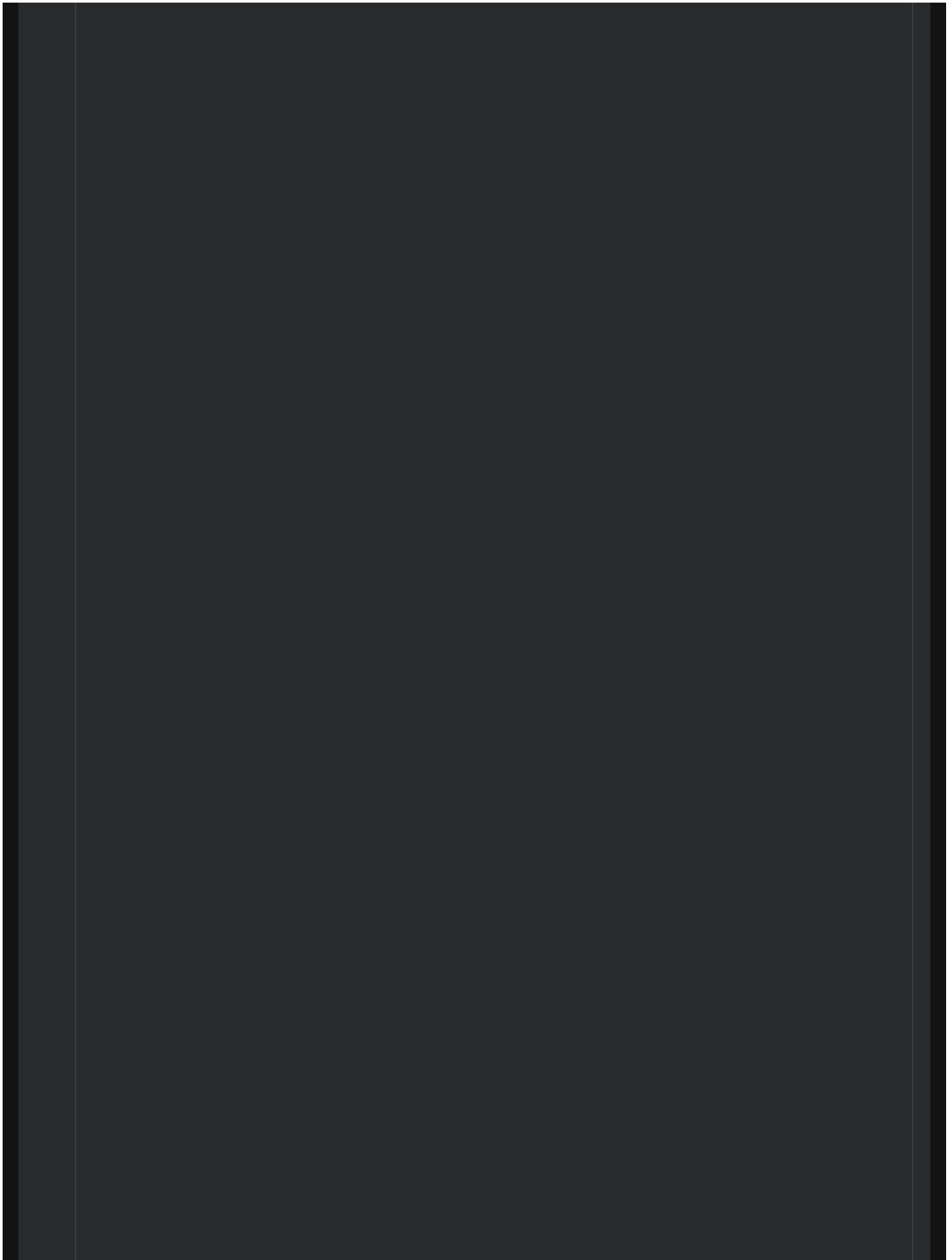


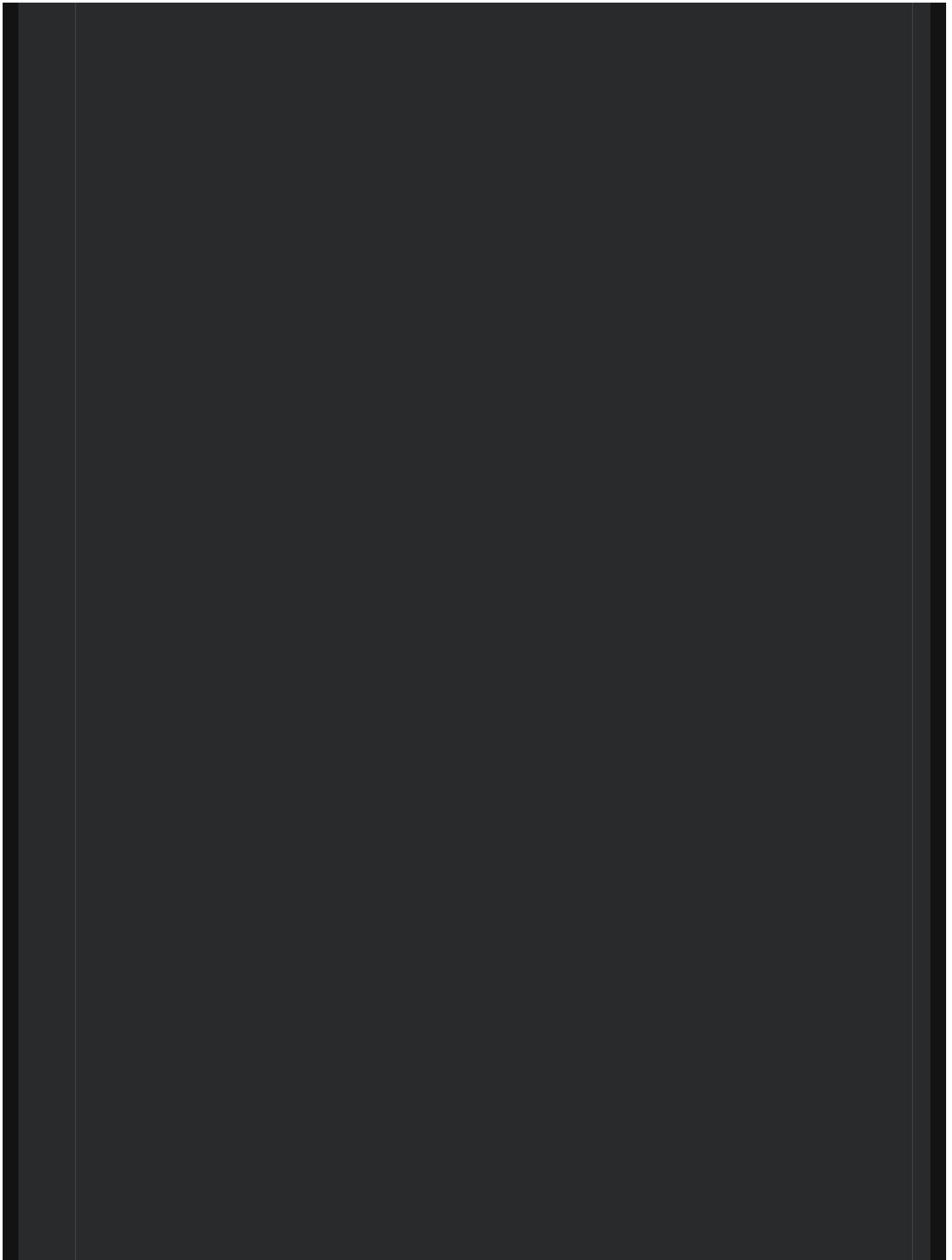






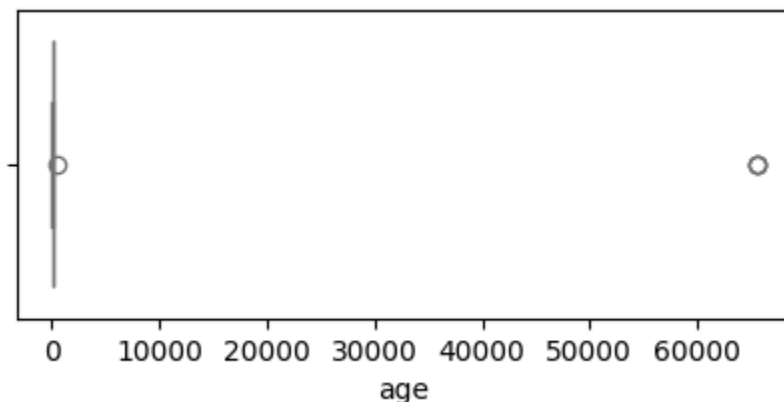


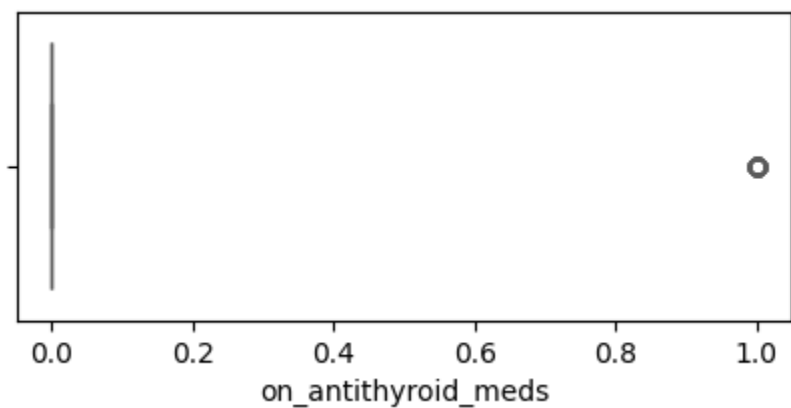
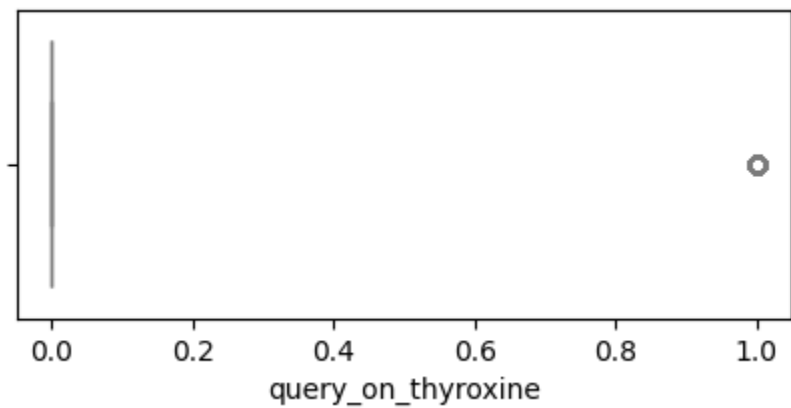
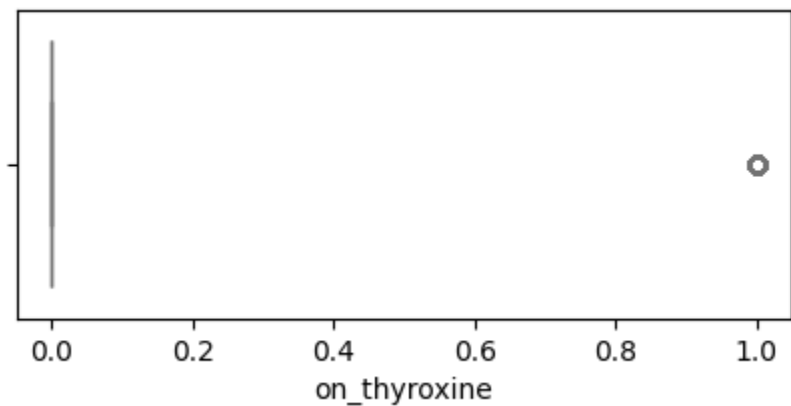
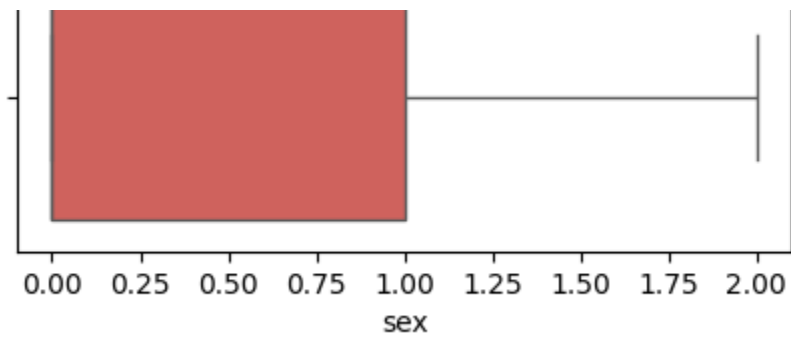


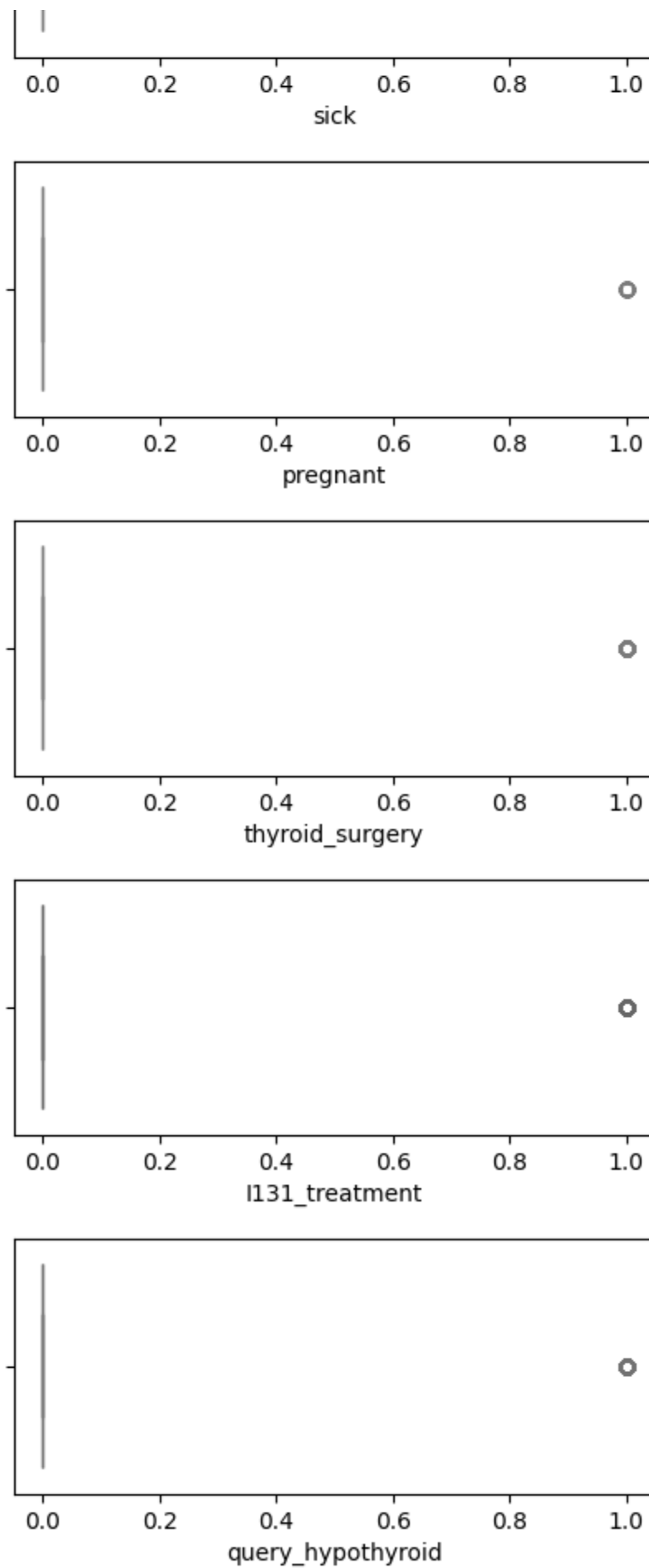


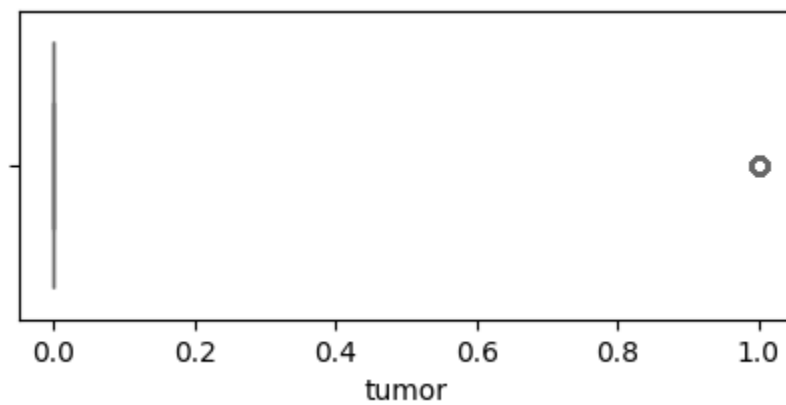
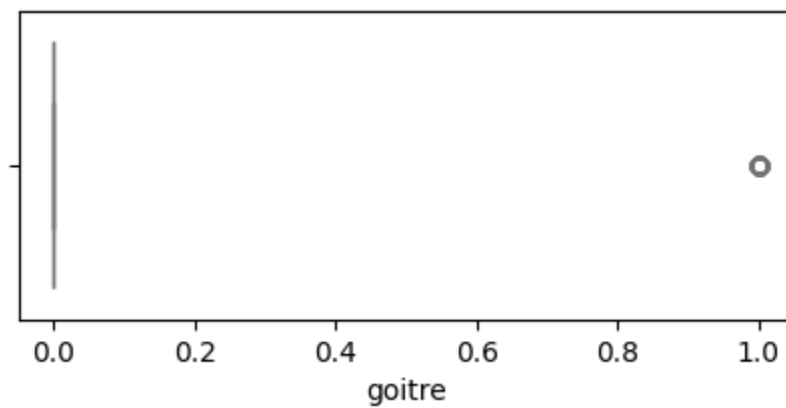
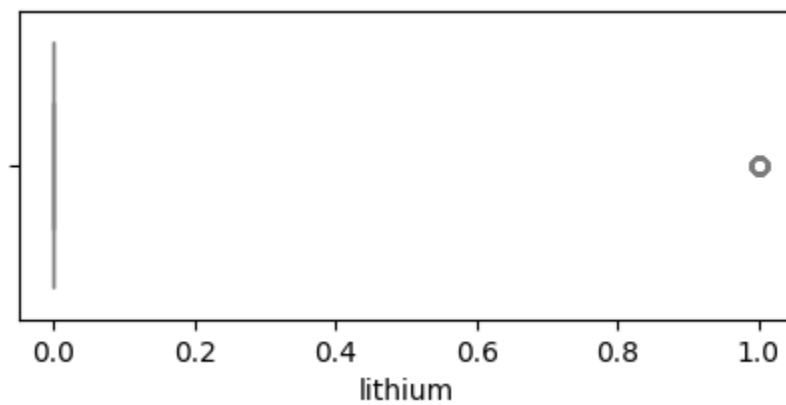
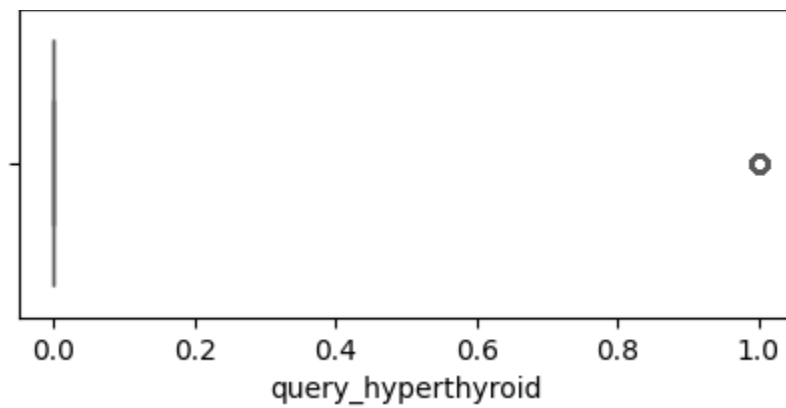
1 Start coding or [generate](#) with AI.

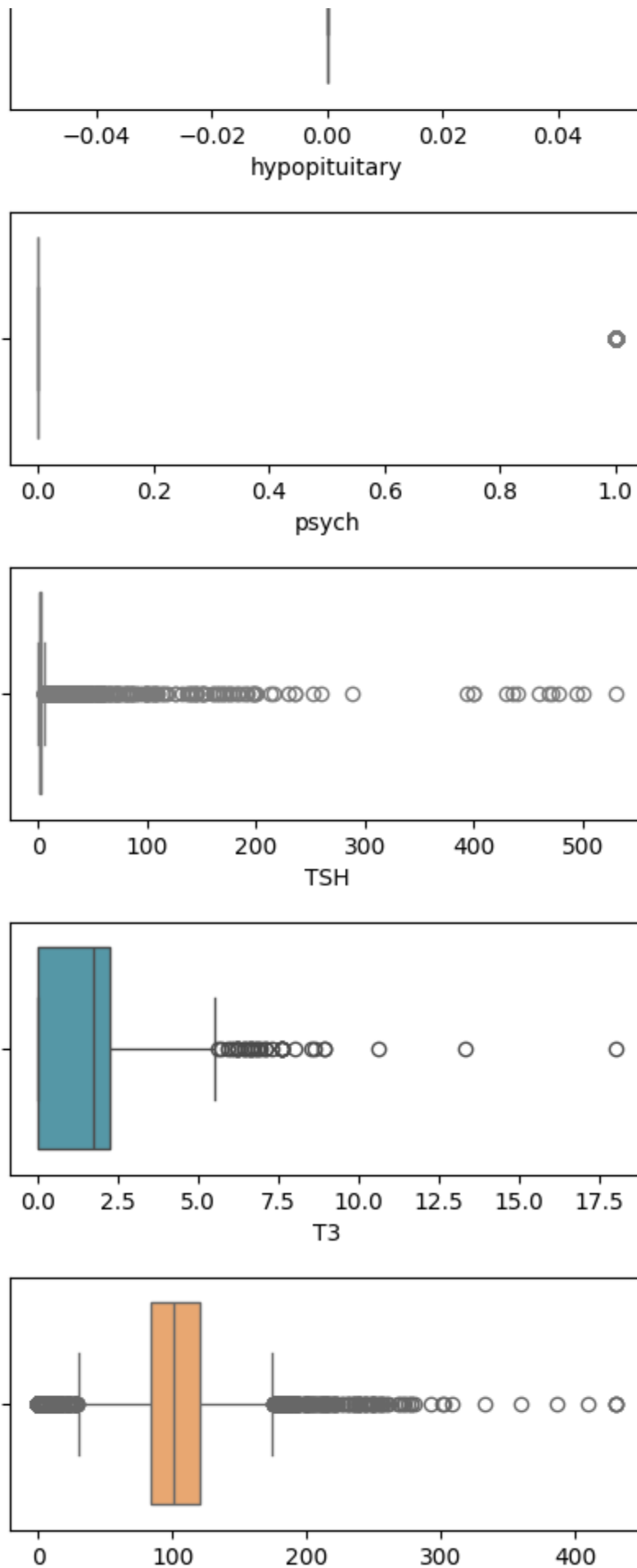
```
1 color=sns.color_palette("Spectral")
2 def boxPlot(dff):
3     d = data_new.drop(columns=['target'])
4     for column in d:
5         plt.figure(figsize=(5, 2))
6         sns.boxplot(x=column, data=d,orient='v',color=random.choice
7
8 boxPlot(data_new)
```

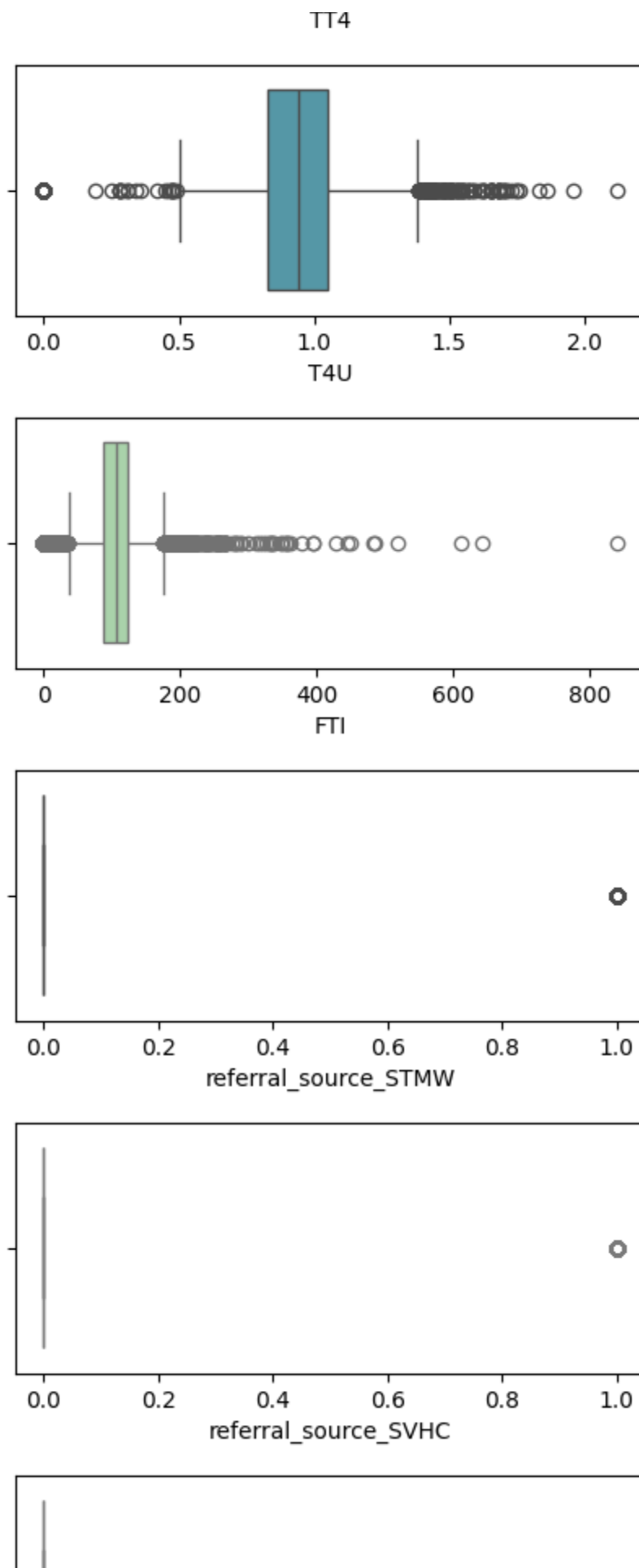


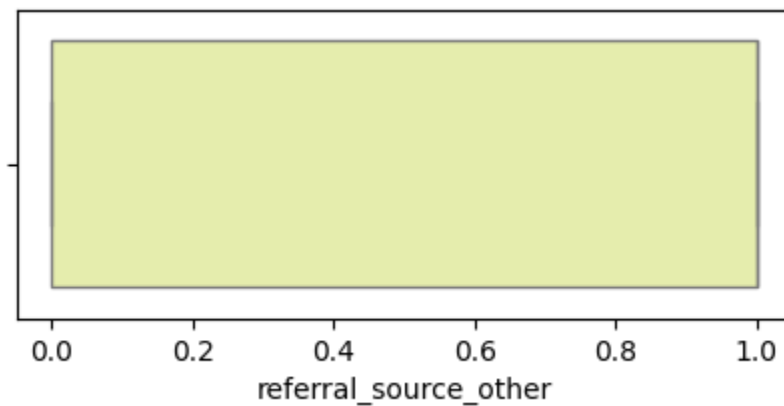
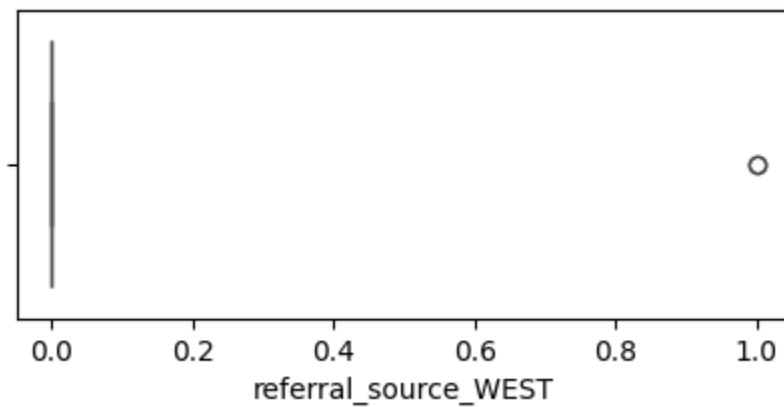
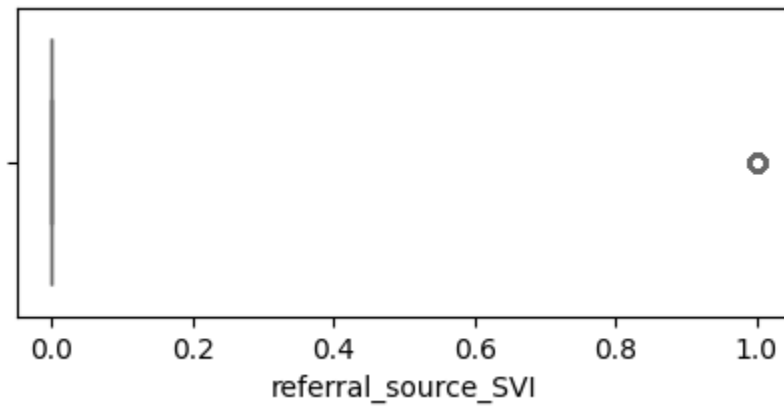
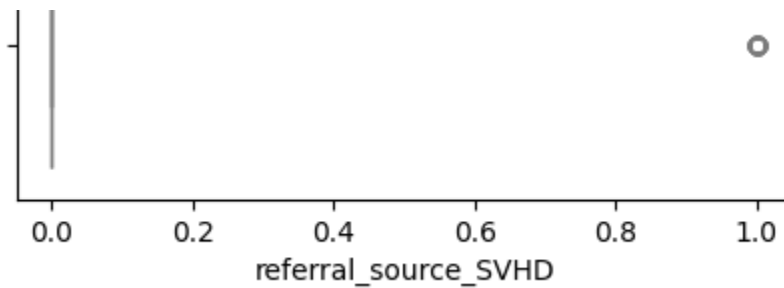










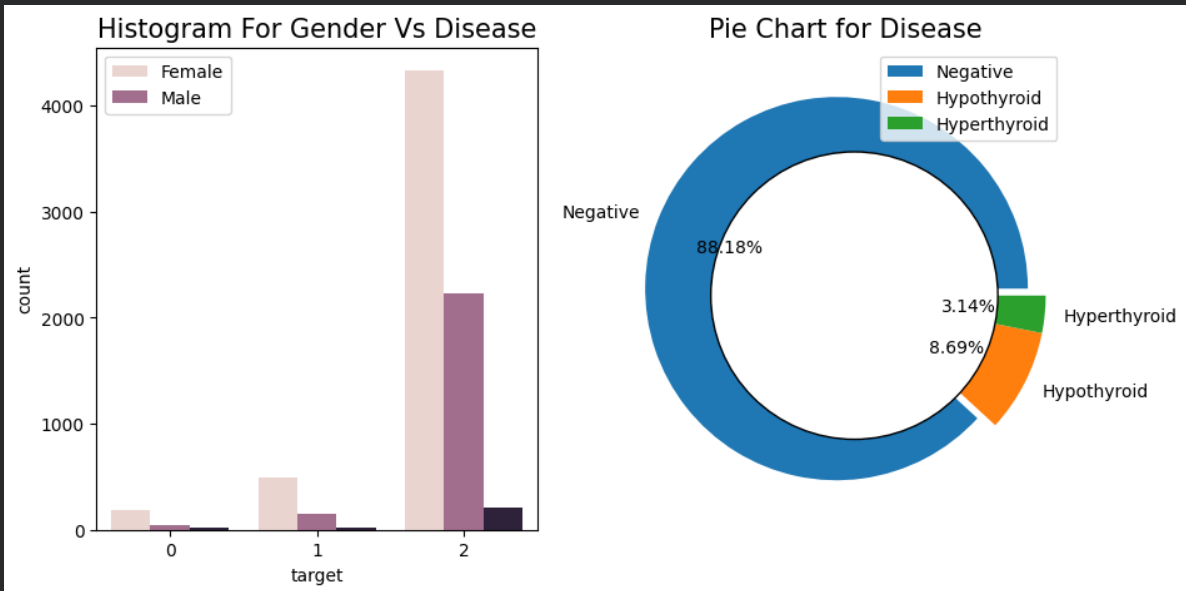


```
1 def configure_plotly_browser_state():
2     import IPython
3     display(IPython.core.display.HTML('''
4         <script src="/static/components/requirejs/require.js"></scr
5         <script>
6             requirejs.config({
7                 paths: {
```

```
7         paths: {
8             base: '/static/base',
9             plotly: 'https://cdn.plot.ly/plotly-latest.min.js?noe
10         },
11     });
12 </script>
13 '''))
```

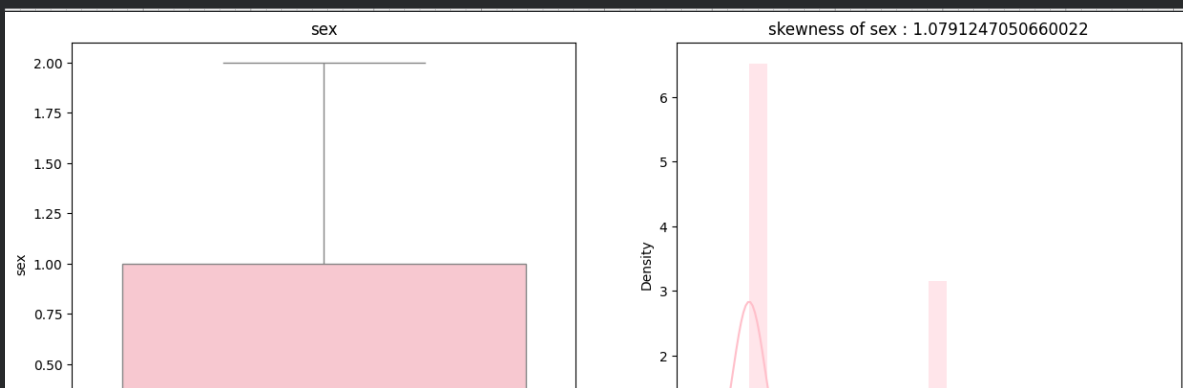
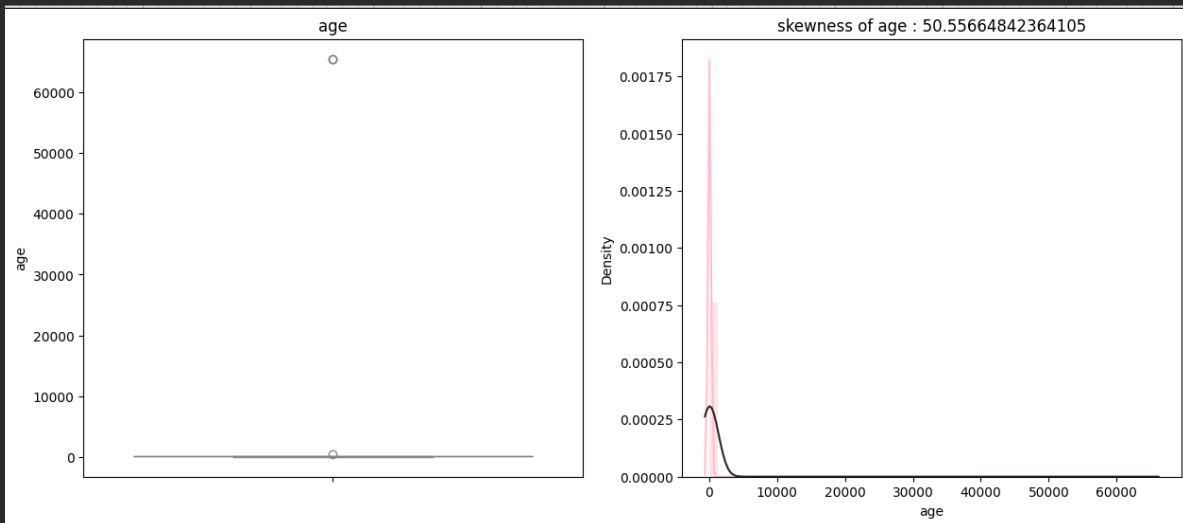
```
1 # Pie chart of target value
2 configure_plotly_browser_state()
3 trace = go.Pie(labels = ['female','male'], values = data_new['sex']
4               textfont=dict(size=10), opacity = 0.9,
5               marker=dict(colors=['hotpink','lightpurple'],
6                           line=dict(color='#000000', width=0.3)))
7
8
9 layout = dict(title = 'Distribution of sex variable')
10
11 fig = dict(data = [trace], layout=layout)
12 py.iplot(fig)
```

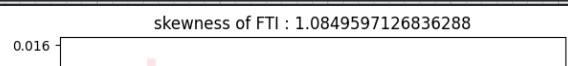
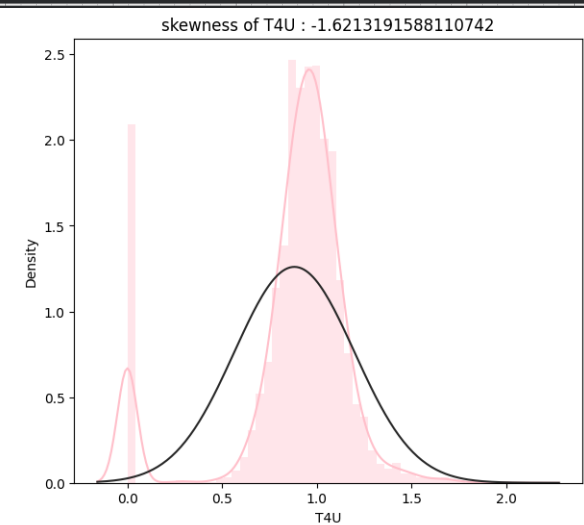
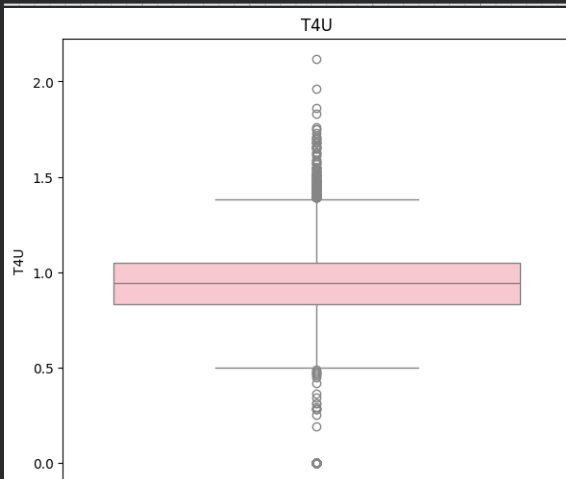
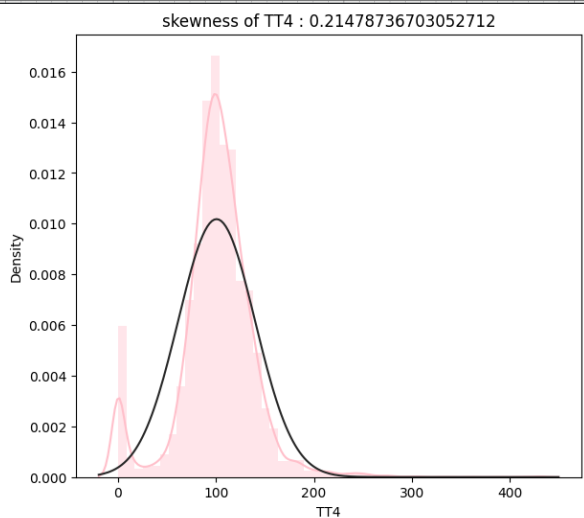
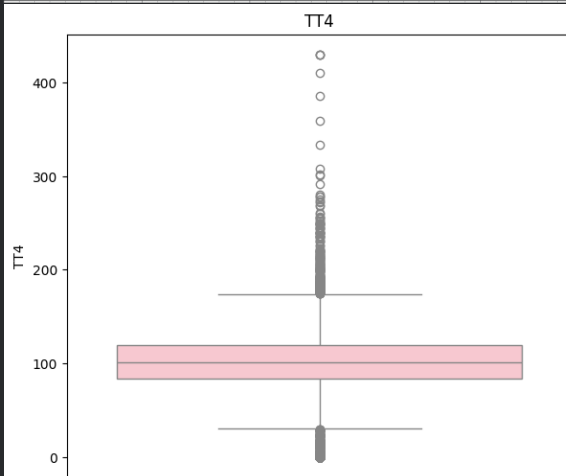
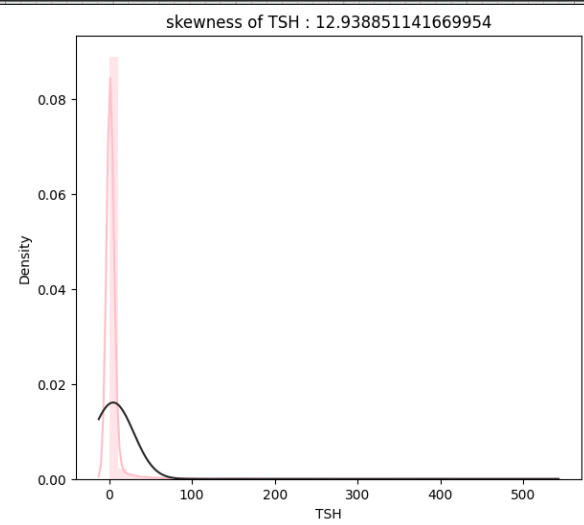
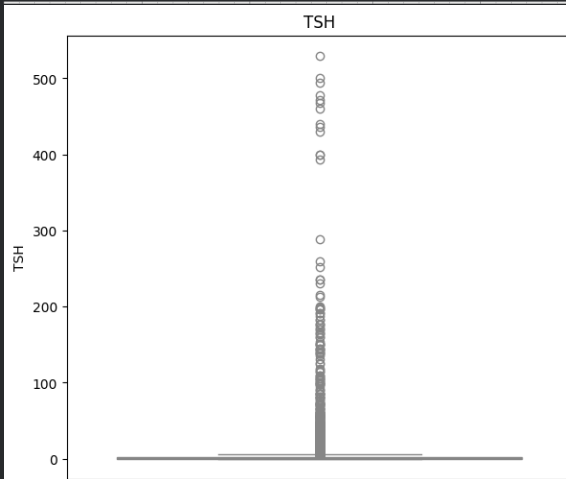
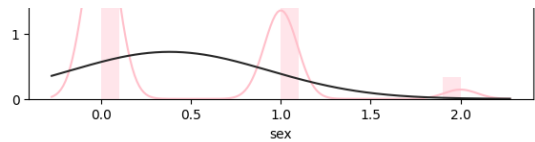
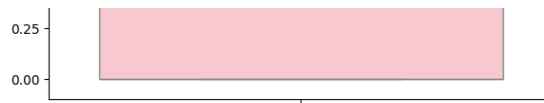
```
1 # Pie chart of target value
2 plt.figure(figsize=(15,5))
3 plt.subplot(1, 2, 1)
4 labels = ['male', 'female']
5 sns.countplot(data = data_new, x='target', hue='sex', alpha=1)
6 plt.legend(labels)
7 plt.title('Histogram For Gender Vs Disease',size=15)
8 configure_plotly_browser_state()
9 labels = ['Negative', 'Hypothyroid','Hyperthyroid']
10 plt.subplot(1, 2, 2)
11 y = [Negative, Hypothyroid, Hyperthyroid]
12 explode = [0, 0.0,0]
13 labels = ['Negative', 'Hypothyroid','Hyperthyroid']
14 plt.pie(y, labels=labels, explode=explode,autopct='% .2f%%')
15 plt.axis('equal')
16 plt.legend(labels)
17 circle = plt.Circle(xy=(0, 0), radius=0.75, facecolor='White', edge
18 plt.gca().add_artist(circle)
19 plt.title('Pie Chart for Disease', size=15)
20 plt.show()
21
```

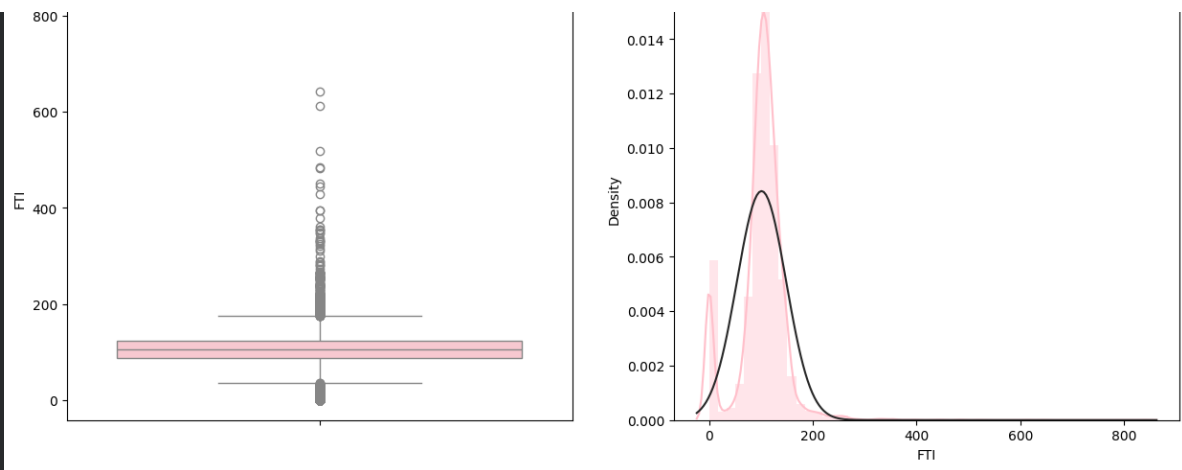


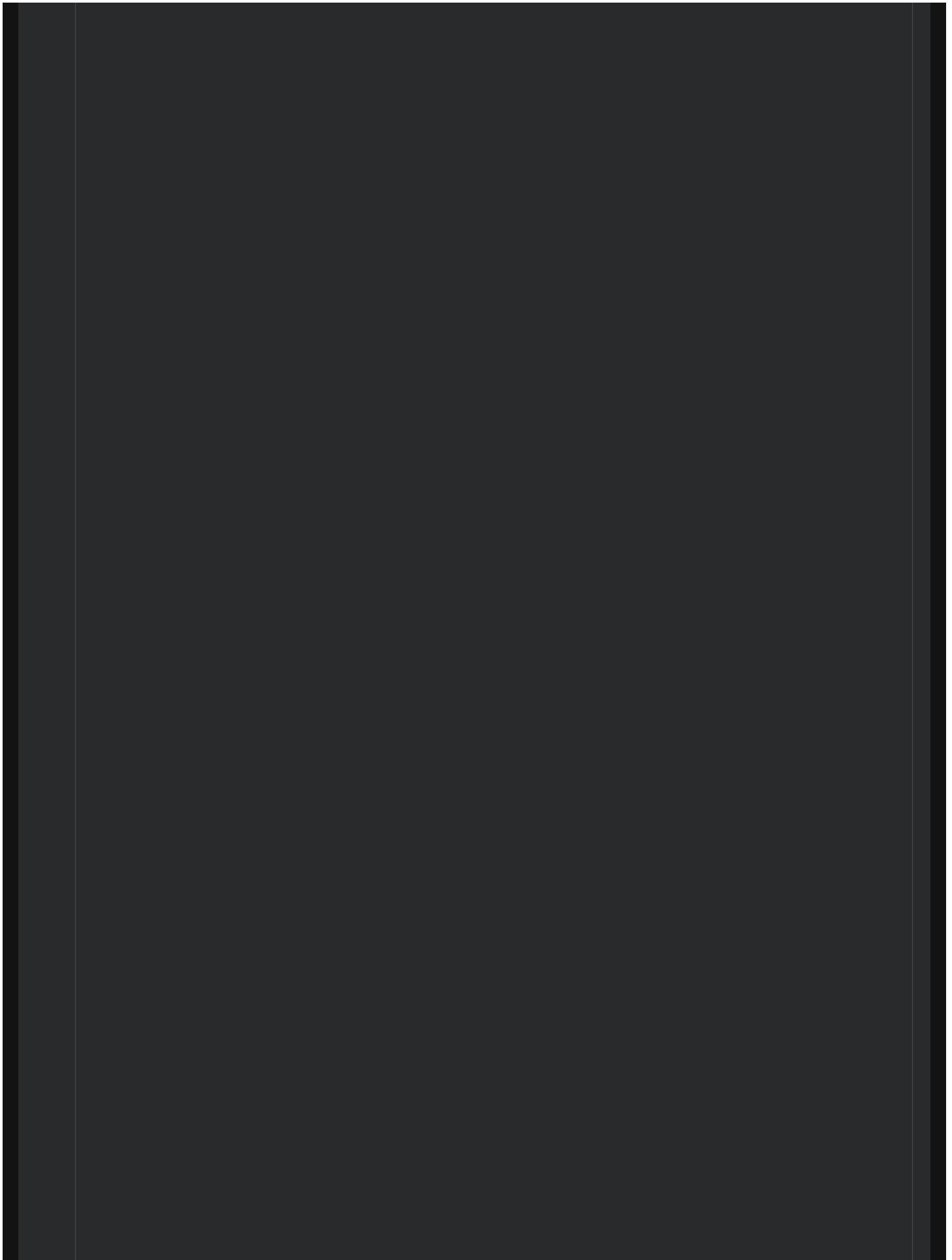
✓ Removing outliers

```
1 from scipy.stats import norm
2
3 # Removing "Date" variable from numerical variable
4 num_var=[var for var in data_new.columns if var in ["TSH", 'TT4',
5
6 # Plotting Box and Distribution plot
7 for var in num_var:
8     plt.figure(figsize=(15,6))
9     plt.subplot(1,2,1)
10    ax=sns.boxplot(data=data_new[var], color = 'pink')
11    ax.set_title(f'{var}')
12    ax.set_ylabel(var)
13
14    plt.subplot(1,2,2)
15    ax=sns.distplot(data_new[var], fit=norm, color = 'pink')
16    ax.set_title(f'skewness of {var} : {data_new[var].skew()}')
17    ax.set_xlabel(var)
18    print('__'*50)
19    plt.show()
```









```

1 # Using Inter Quartile Range for removing outliers from numerical v
2 from scipy.stats import norm
3 # Defining outlier features
4 outlier_var= num_var
5
6 # Capping dataset
7 for i in outlier_var:
8     # Findling IQR
9     Q1 = data_new[i].quantile(0.25)
10    Q3 = data_new[i].quantile(0.75)
11    IQR = Q3-Q1
12
13    # Defining upper and lower limit
14    lower_limit = data_new[i].quantile(0.25)-1.5*IQR
15    upper_limit = data_new[i].quantile(0.75)+1.5*IQR
16
17    # Applying lower and upper limit to each variables
18    data_new.loc[(data_new[i] > upper_limit),i] = upper_limit
19    data_new.loc[(data_new[i] < lower_limit),i] = lower_limit

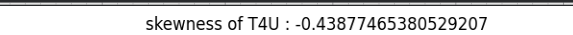
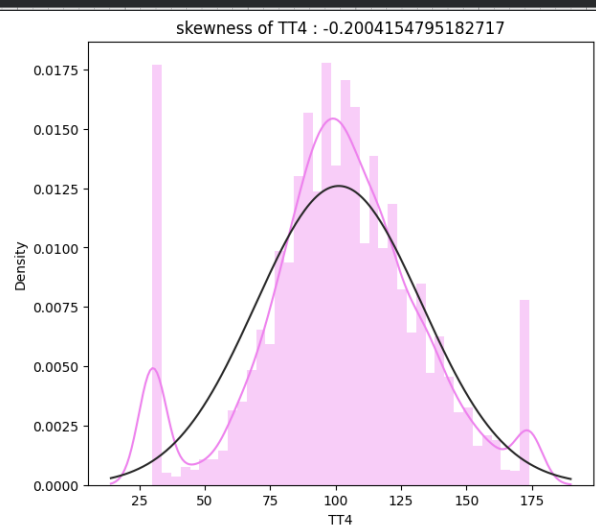
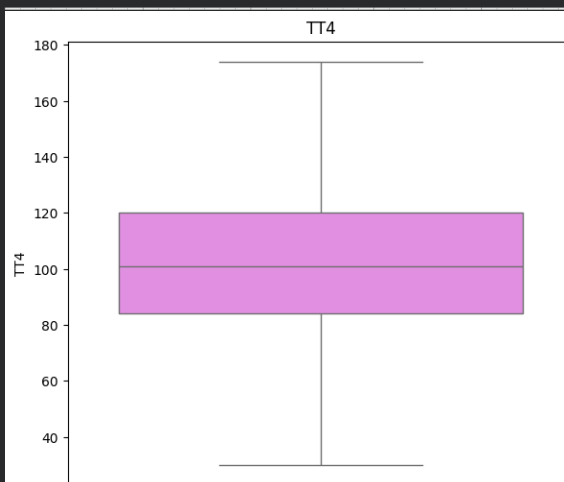
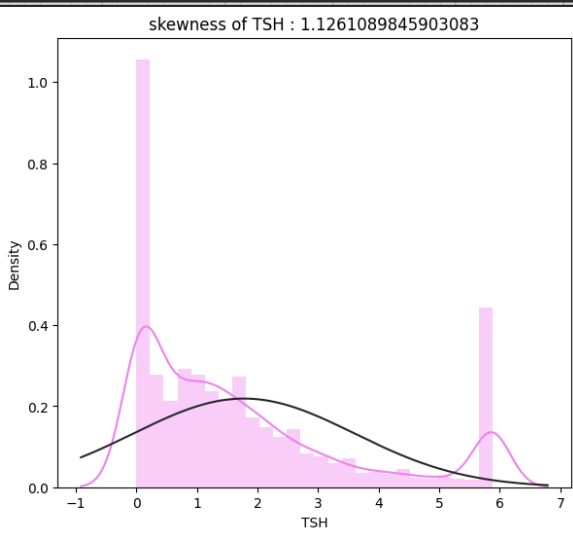
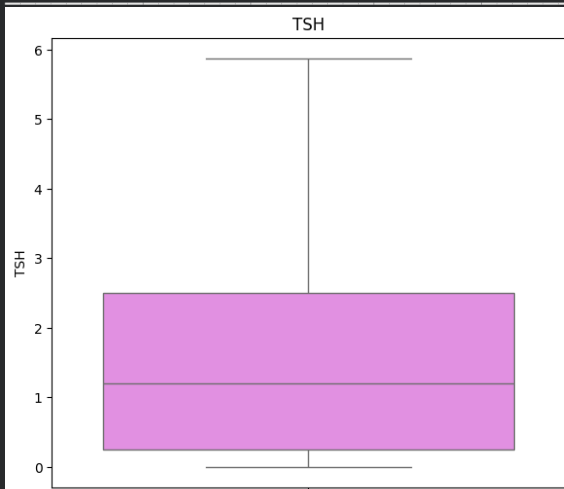
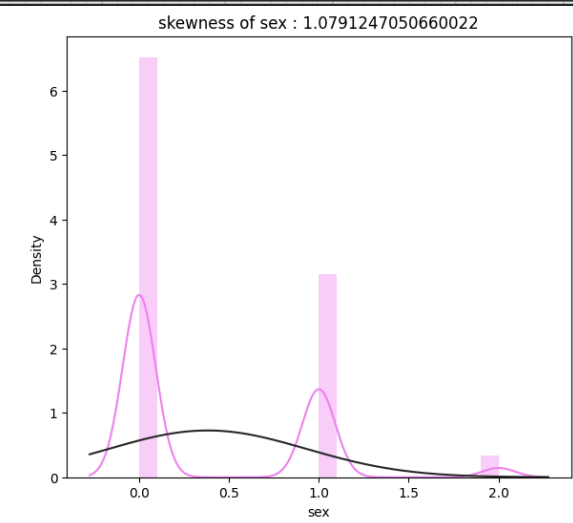
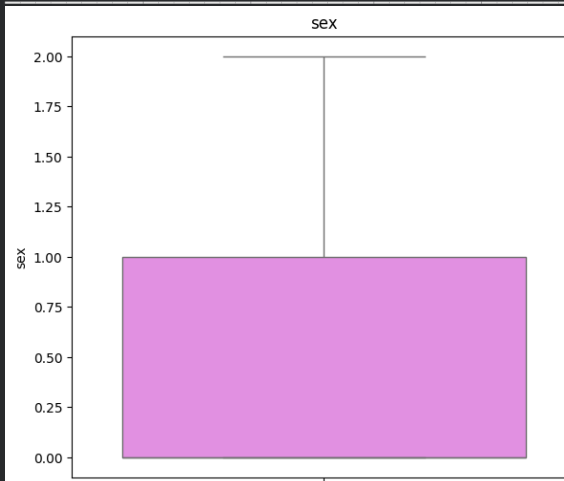
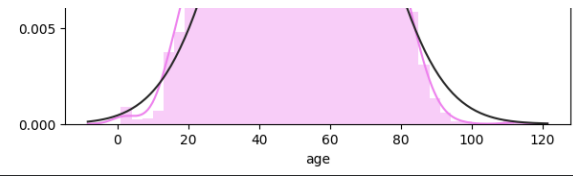
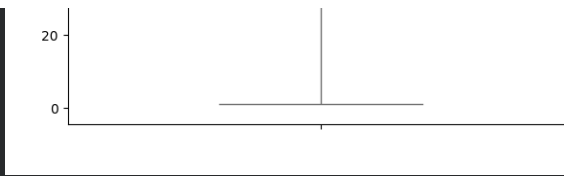
```

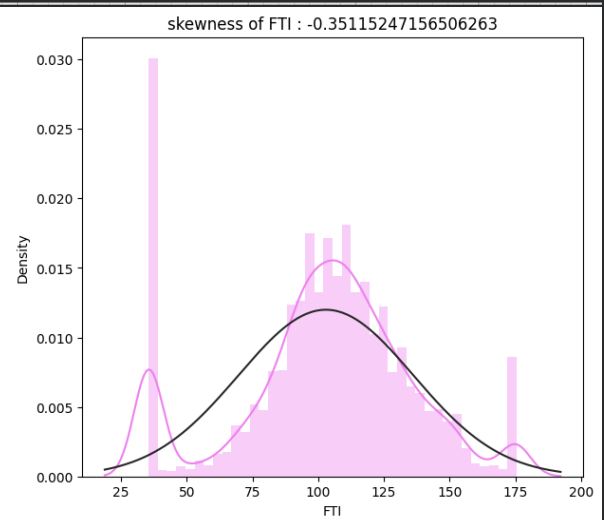
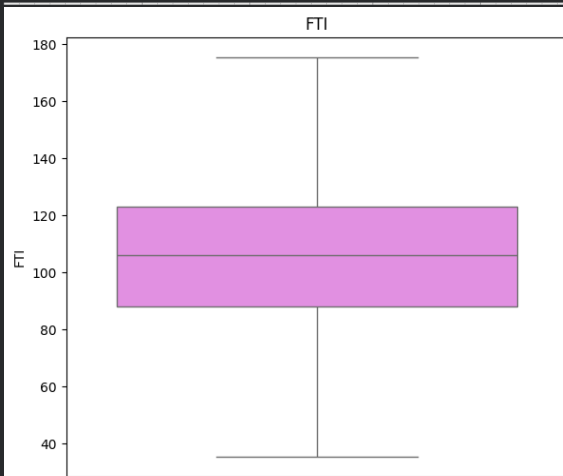
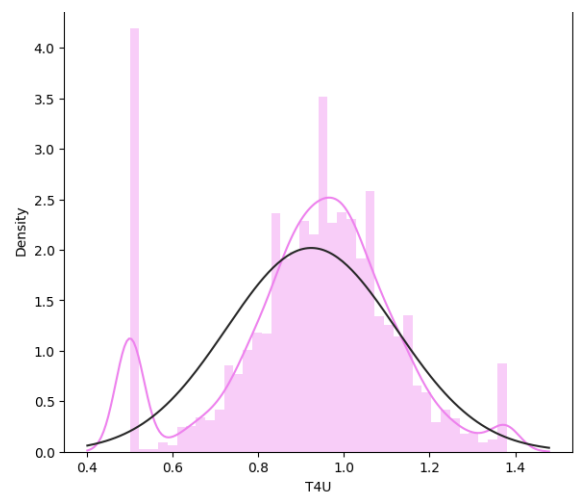
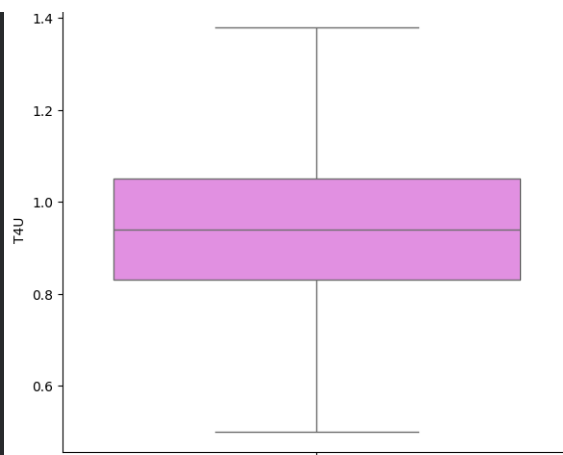
```

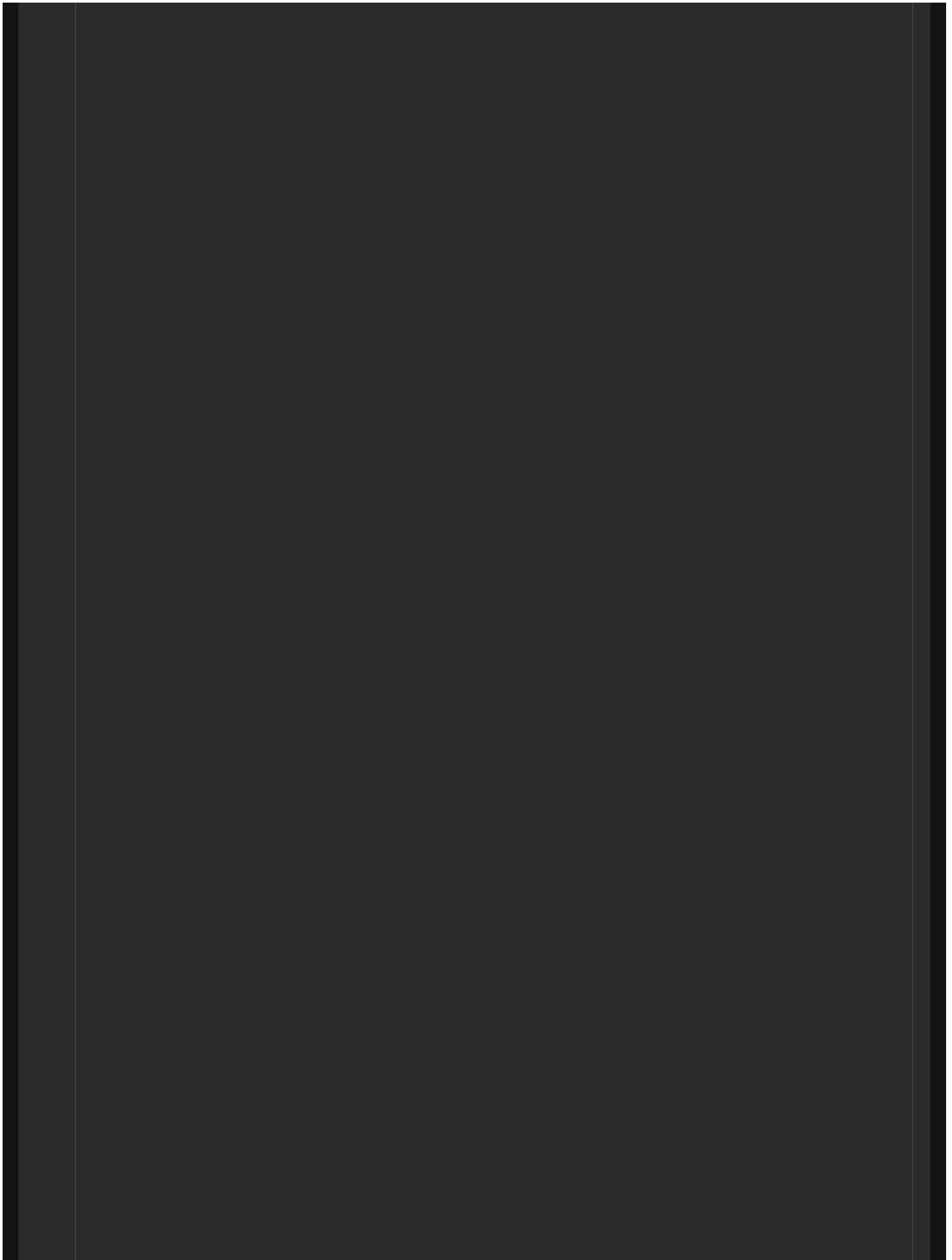
1 from scipy.stats import norm
2 for var in outlier_var:
3     plt.figure(figsize=(15,6))
4     plt.subplot(1,2,1)
5     ax=sns.boxplot(data=data_new[var], color = 'violet')
6     ax.set_title(f'{var}')
7     ax.set_ylabel(var)
8
9     plt.subplot(1,2,2)
10    ax=sns.distplot(data_new[var], fit=norm, color = 'violet')
11    ax.set_title(f'skewness of {var} : {data_new[var].skew()}')
12    ax.set_xlabel(var)
13    print('__'*50)
14    plt.show()

```



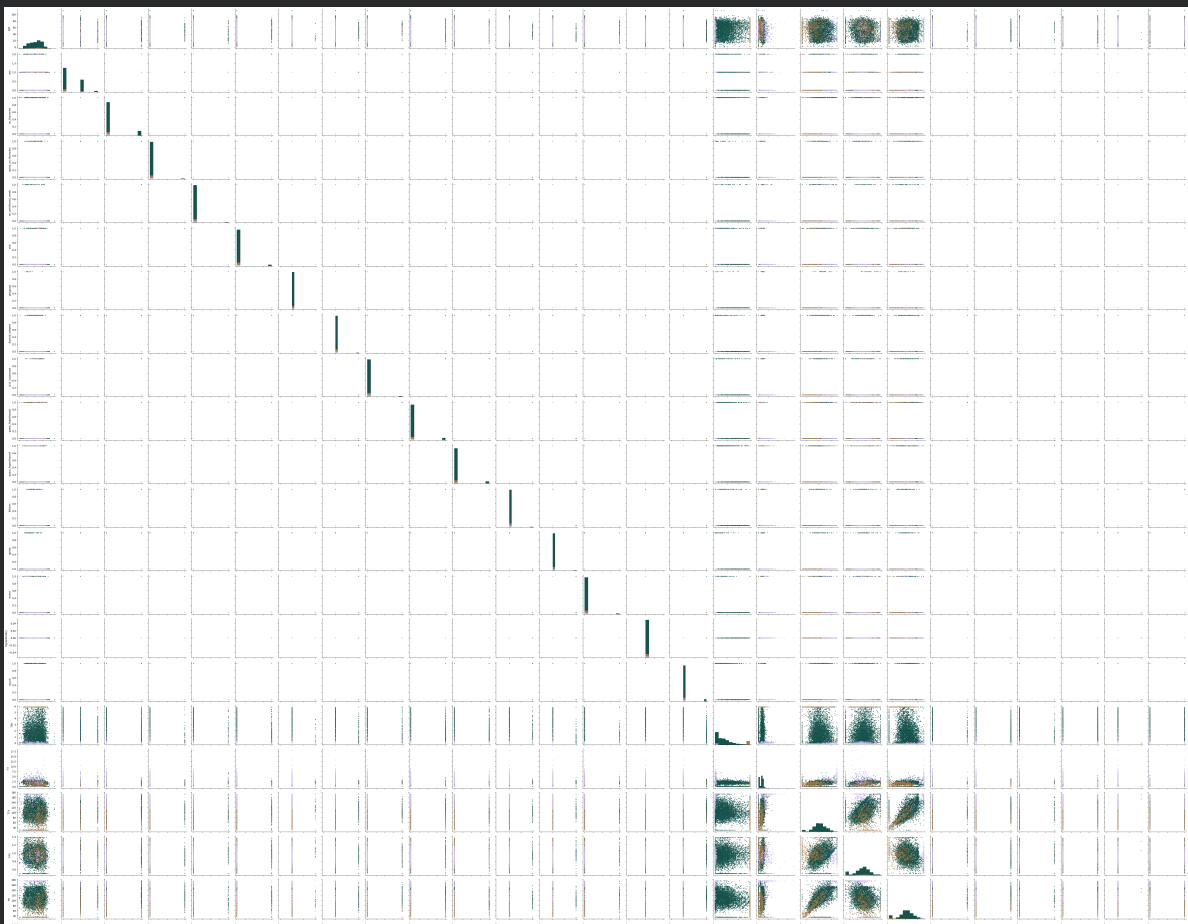






✓ correlation

```
1 g = sns.PairGrid(data_new, hue='target', palette='cubehelix')
2 g = g.map_diag(plt.hist)
3 g = g.map_offdiag(plt.scatter, s = 3)
```






```
1 data_new['target'].unique()
```

```
array([2, 1, 0])
```

✓ Encoding categorical target value

```
1 from sklearn.preprocessing import LabelEncoder
2 le = LabelEncoder()
3 data_new['target']=le.fit_transform(data_new['target'])
```

```
1 #correlation
2 correlation = data_new.corr()
3 #tick labels
4 matrix_cols = correlation.columns.tolist()
5 #convert to array
6 corr_array = np.array(correlation)
7 pd.DataFrame(corr_array)
```

	0	1	2	3	4	5	6	
0	1.000000	-0.028978	-0.000436	-0.012784	-0.051342	0.068341	-0.080568	-0.016158
1	-0.028978	1.000000	-0.102551	0.034257	-0.028411	0.008345	-0.036991	-0.031231
2	-0.000436	-0.102551	1.000000	0.004579	-0.012145	-0.036693	0.009066	0.025850
3	-0.012784	0.034257	0.004579	1.000000	-0.014050	-0.008734	0.005177	0.002686
4	-0.051342	-0.028411	-0.012145	-0.014050	1.000000	-0.021698	0.008659	-0.002769
5	0.068341	0.008345	-0.036693	-0.008734	-0.021698	1.000000	-0.014258	-0.005659
6	-0.080568	-0.036991	0.009066	0.005177	0.008659	-0.014258	1.000000	-0.008561
7	-0.016158	-0.031231	0.025850	0.002686	-0.002769	-0.005659	-0.008561	1.000000
8	0.055752	-0.010076	0.069228	-0.017450	-0.015061	-0.026949	-0.009897	0.009742
9	0.037308	-0.082124	0.091583	-0.017873	-0.024914	0.009742	-0.019502	-0.028935
10	-0.041772	-0.071660	-0.038436	-0.027846	0.097873	-0.028935	0.035299	0.007701
11	-0.033409	0.025316	0.005840	-0.013578	-0.011719	-0.007922	-0.007701	-0.019437
12	-0.070025	-0.019860	0.003436	0.008650	-0.010862	-0.019437	0.066933	0.145947
13	-0.042412	-0.077293	-0.008979	-0.008093	-0.018146	0.001767	0.145947	0.000820
14	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN
15	-0.087840	0.080620	-0.053851	-0.029152	-0.025161	-0.032375	-0.008203	-0.000820

16	0.077427	-0.021678	-0.071577	-0.068321	-0.015666	-0.001162	-0.038119	0.
17	-0.048161	0.039067	-0.067959	-0.059525	0.088007	-0.031890	0.046710	-0.
18	0.011917	-0.072792	0.169397	0.003472	0.018957	-0.014070	0.057596	0.
19	-0.012859	-0.130271	0.074081	-0.012131	0.017655	0.005224	0.098039	0.
20	0.080162	0.019602	0.138016	0.033527	-0.008480	-0.013174	-0.000760	0.
21	-0.015564	0.046572	0.084921	0.015240	-0.011627	0.020386	-0.058164	0.
22	-0.156237	-0.039618	0.010186	-0.018009	0.001755	-0.027812	0.277370	0.
23	-0.115989	0.140983	-0.091864	-0.045547	-0.039311	-0.061669	-0.020119	-0.
24	0.028819	-0.007683	0.006404	0.012240	-0.009772	0.263226	-0.006422	0.
25	0.251526	0.053095	-0.167012	-0.034825	-0.060457	0.053801	-0.041550	-0.
26	-0.016007	-0.011188	0.045750	-0.002059	-0.001777	-0.003180	-0.001168	-0.
27	-0.107801	-0.125614	0.202071	0.063410	0.080422	-0.047149	-0.027673	0.

```
1 #Heat map plot
2 configure_plotly_browser_state()
3 trace = go.Heatmap(z = corr_array,x = matrix_cols,y = matrix_cols,x
4 layout = go.Layout(dict(title = 'Correlation Matrix for variables',
5 t = 25,b = 210),yaxis = dict(tickfont = dict(size = 9)),xaxis =
6 fig = go.Figure(data = [trace],layout = layout)
7 py.iplot(fig)
```

▼ Train_Test Model

```
1 random_state = 18
2 X = data_new.drop(columns='target')
3 y = data_new['target']
4 #Stratified Split:
5 X_train, X_test, y_train, y_test = train_test_split(X, y, test_size
6 X_train, X_val, y_train, y_val = train_test_split(X_train, y_train,
```

```
1 def prepare_model(algorithm):
2     model = algorithm
3     model.fit(X_train , y_train)
4     return model
```

```
1 def prepare_confusion_matrix(algorithm, model):
2     print(algorithm)
3     plt.figure(figsize=(10,5))
4     pred = model.predict(X_test)
5     cm = confusion_matrix(y_test, pred)
6     ax= plt.subplot()
7     sns.heatmap(cm, annot=True, fmt='g', ax=ax)
8     plt.show()
9
10    # labels, title and ticks
11    ax.set_xlabel('Predicted labels');ax.set_ylabel('True labels');
12    ax.set_title('Confusion Matrix');
```

```

12 ax.set_title(' Confusion Matrix ');
13

```

```

1 def prepare_pred_plot(algorithms, model):
2     print(algorithms)
3     plt.figure(figsize=(10, 10))
4     plt.plot(model_and_score_test, label='Predicted')
5     plt.plot(y_test.values, label='Actual')
6     plt.ylabel(algorithms)
7     plt.legend()
8     plt.show()

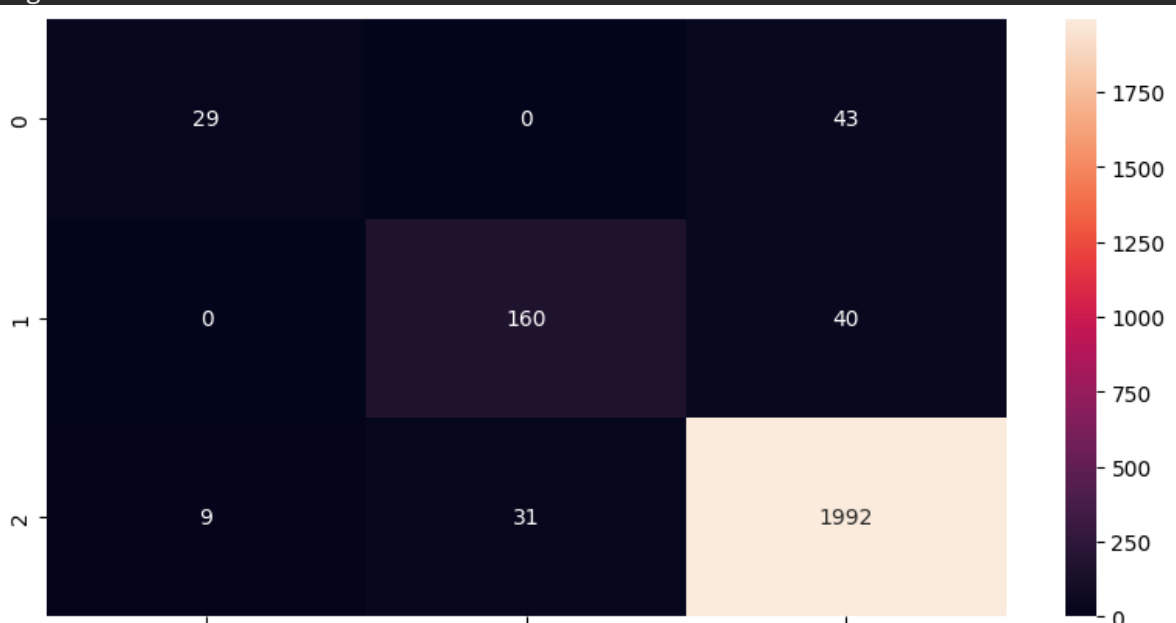
```

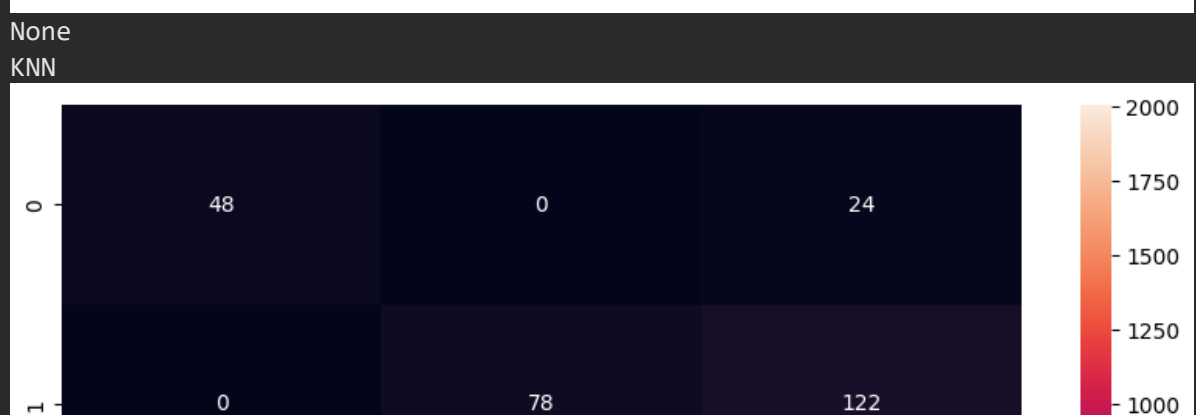
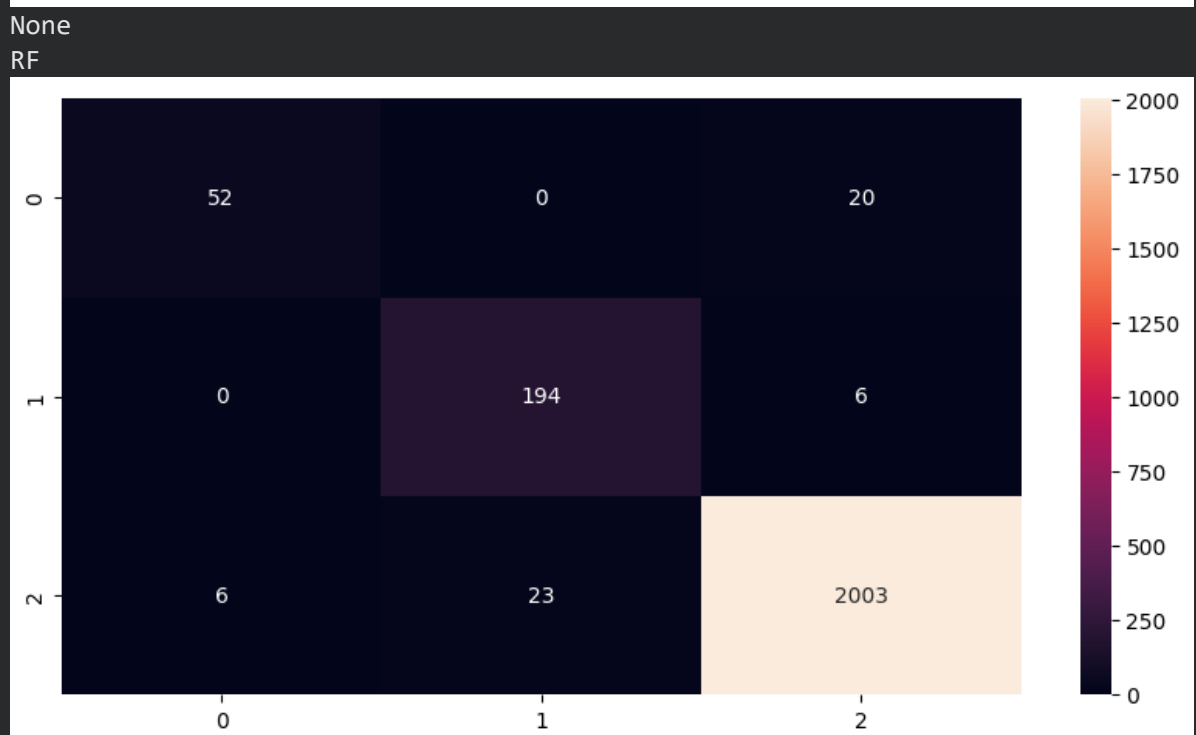
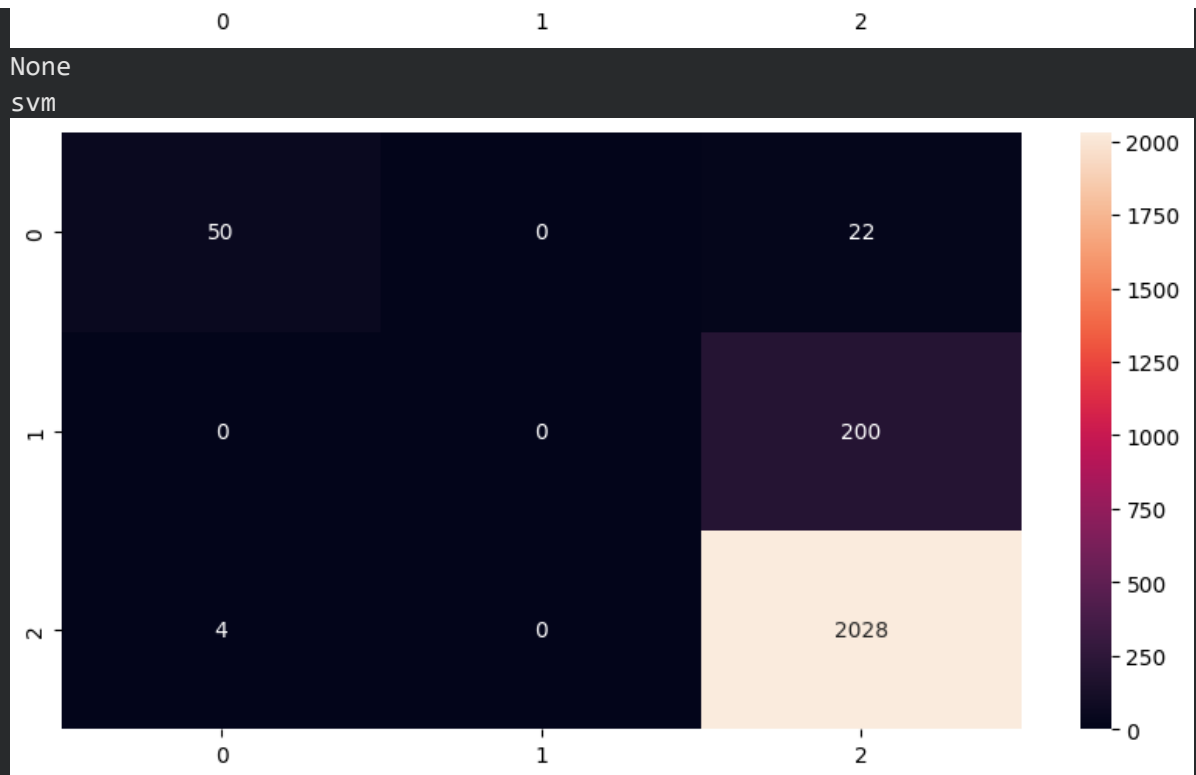
```

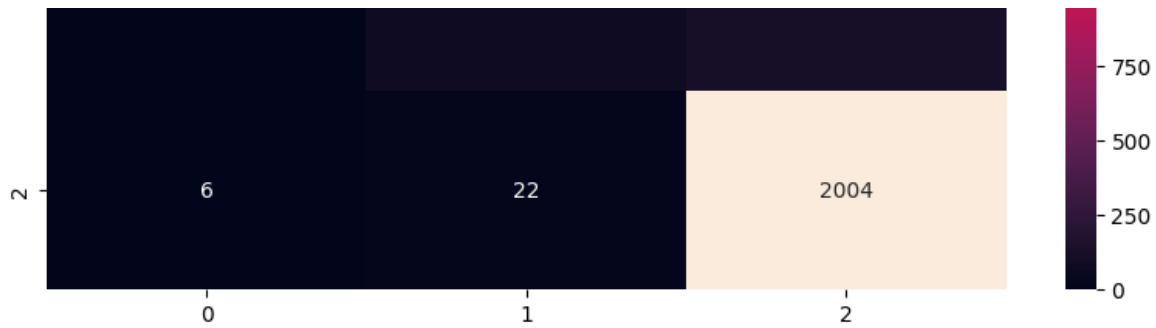
1 algorithms = [('log', LogisticRegression(random_state=18)),
2               ('svm', SVC(random_state=18)),
3               ('RF', RandomForestClassifier(random_state=18)),
4               ('KNN', KNeighborsClassifier()),
5               ('NB', GaussianNB()),
6               ('DT', DecisionTreeClassifier(random_state = 18))]
7 trained_models = []
8 model_and_score_valid = {}
9 model_and_score_test = {}
10
11 for index, tup in enumerate(algorithms):
12     model = prepare_model(tup[1])
13     model_and_score_valid[tup[0]] = str(model.score(X_val,y_val)*100)
14     model_and_score_test[tup[0]] = str(model.score(X_test,y_test)*100)
15     trained_models.append((tup[0],model))
16     print(prepare_confusion_matrix(tup[0], tup[1]))
17 print(model_and_score_valid)
18 print(model_and_score_test)

```

log

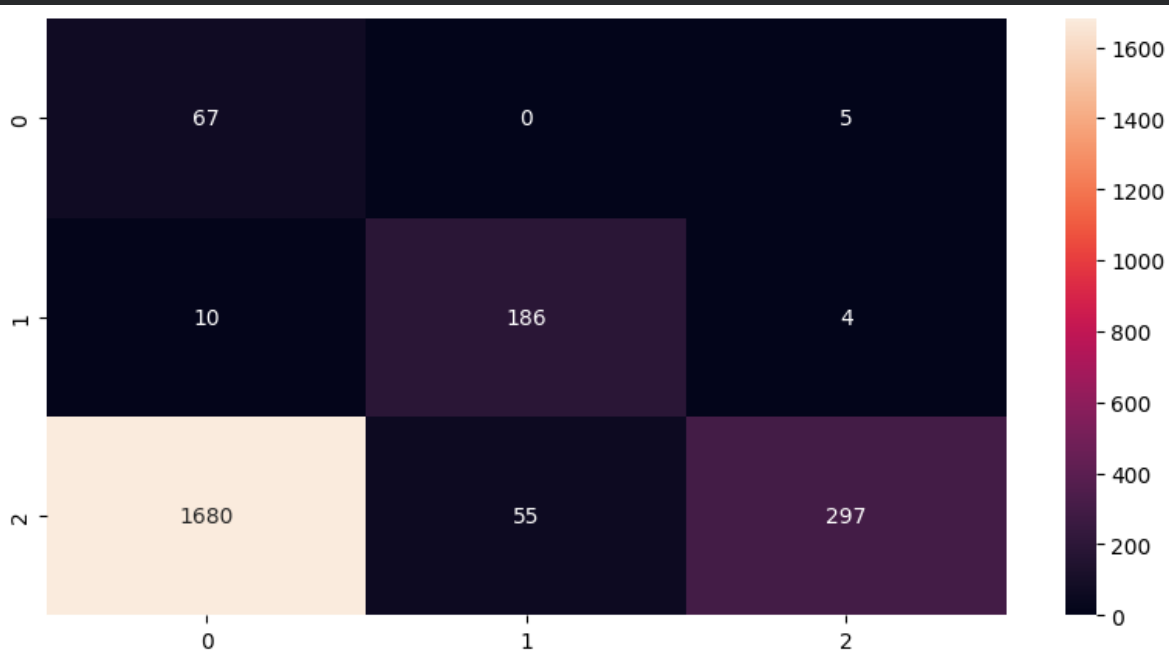






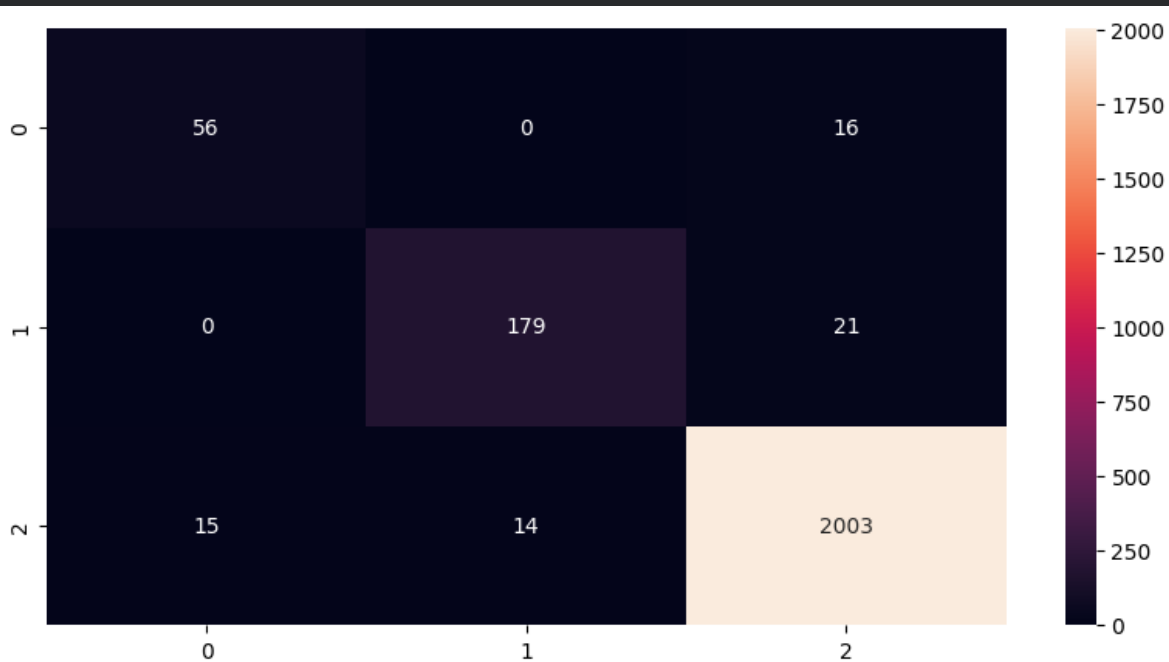
None

NB



None

DT



None

```
{ 'log': '94.04761904761905%', 'svm': '89.95535714285714%', 'RF': '97.87946428571429%' }  
{ 'log': '94.66145833333334%', 'svm': '90.19097222222222%', 'RF': '97.61284722222222%' }
```

